

Learning localized visual representations from radiology reports for chest X-ray vision–language models

Kang Liu^{1,2,3,†}, Zhuoqi Ma^{1,2,3,†}, Kun Xie^{1,2,3}, Yunan Li^{1,2,3}, Ruyi Liu^{1,2,3}, Ronghan Li^{1,2,3}, Guang Jia^{1,2}, and Qiguang Miao^{1,2,3,*}

¹School of Computer Science and Technology, Xidian University

²Xi'an Key Laboratory of Big Data and Intelligent Vision

³Key Laboratory of Smart Education of Shaanxi Higher Education Institutes, Xidian University

[†]These authors contributed equally to this work.

^{*}Corresponding author: qgmiao@xidian.edu.cn

Abstract

Medical vision–language models have advanced chest X-ray (CXR) interpretation, yet many predominantly learn global visual representations, which may be insufficient to capture localized pathological patterns. Existing approaches to localized representation learning often rely on spatial annotations, such as bounding boxes or masks, which are costly to obtain at scale. Here we introduce **CheXLoc**, a vision–language framework that learns localized visual representations directly from paired CXRs and reports without spatial supervision. CheXLoc treats clinically meaningful entities in radiology reports as weak supervisory signals and combines global image–report alignment with entity-guided local patch–token alignment. Pretrained on 240,197 CXRs from 59,693 patients in the MIMIC-CXR training set, CheXLoc achieved strong report-generation performance across internal and external cohorts while using substantially fewer parameters and less GPU memory than multimodal foundation models. The learned representations also transferred to disease classification and lesion segmentation, with AUROCs of 0.966 and 0.937 on Shenzhen and TBX11K, respectively, using only 1% of the downstream training data. Image-removal and lesion-level perturbation further indicated greater sensitivity of CheXLoc to visual and disease-specific evidence during report generation. These findings provide evidence that report supervision can serve as a source of weak supervision for learning localized visual representations without requiring spatial annotations.

1 Introduction

Chest X-ray (CXR) imaging is among the most widely used diagnostic modalities in clinical practice, providing critical information for detecting and monitoring thoracic abnormalities [1, 2, 3]. Recent advances in vision–language models (VLMs) have enabled joint learning from large-scale collections of paired radiographs and reports, supporting a range of medical imaging tasks, including disease classification [4, 5], lesion segmentation [6, 7], and report generation [8, 9]. However, existing medical VLMs [10, 11, 12, 13, 14] predominantly learn global image representations that summarize the radiograph as a whole, potentially limiting their ability to preserve localized pathological information.

Several approaches have sought to learn localized visual representations using explicit spatial supervision, including lesion bounding boxes, segmentation masks, or radiologist eye-tracking data. For example, DCXFM [15], EGMA [16], and CoGaze [7] incorporate lesion annotations or radiologist eye-tracking signals to guide visual attention. Although such supervision provides direct spatial constraints, obtaining these annotations requires substantial expert effort and remains difficult at scale. This limits the application of spatially supervised localization methods to large-scale image–report datasets for which corresponding spatial annotations are unavailable.

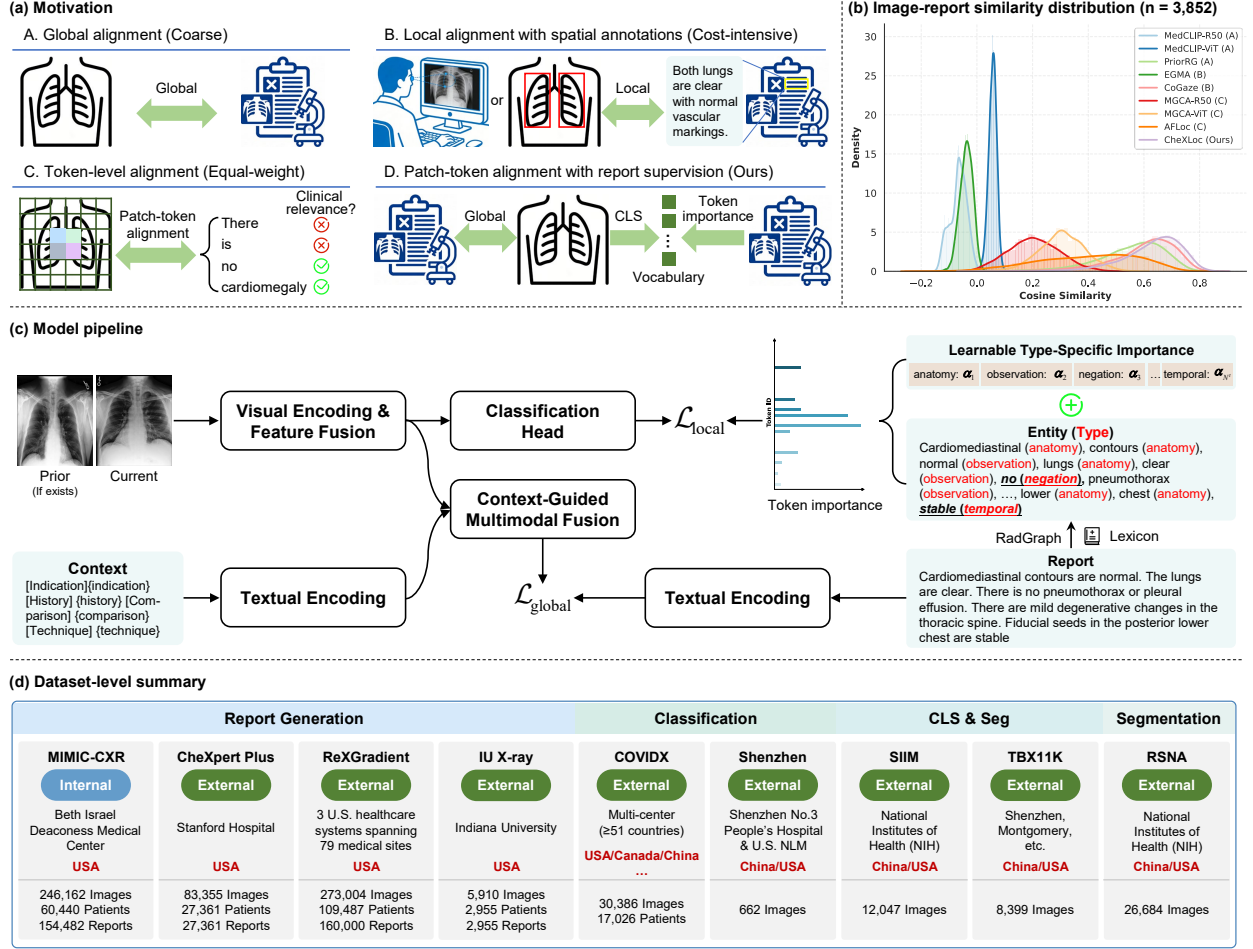
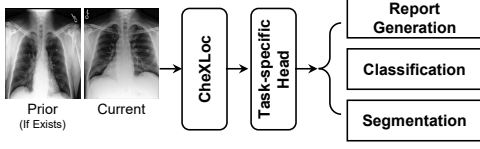


Figure 1: **Overview of CheXLoc pretraining.** **a**, Comparison of existing approaches with CheXLoc. CheXLoc uses entity-guided token importance for local patch-token alignment without spatial annotations, whereas existing methods rely on uniform token weighting or spatial supervision. **b**, Image-report cosine similarity on the full MIMIC-CXR test set ($n = 3,852$). CheXLoc achieves the highest cosine similarity among the evaluated methods despite the absence of spatial supervision. **c**, Architecture of CheXLoc. Italicized, bolded, and underlined entities denote entities omitted by RadGraph and recovered using the expert-curated lexicon. **d**, Dataset statistics for the study cohorts. Reported numbers represent the total size of each dataset. MIMIC-CXR comprises 246,162 images, of which 240,197 images from the official training split were used for pretraining.

An alternative direction is to learn localized visual representations directly from paired radiographs and reports. Recent studies have explored localized cross-modal correspondence through patch-token alignment [17] and multi-level representation alignment [5]. MGCA [17] uses multi-granularity contrastive learning to align visual and textual representations at different semantic levels, while AFLoc [5] investigates hierarchical alignment between visual and textual features. These studies provide evidence that image-report pairs can provide supervision for learning localized cross-modal associations without explicit spatial annotations. However, their alignment objectives do not explicitly distinguish textual tokens according to their relevance to localized visual evidence, treating clinically meaningful entities (e.g., *opacity* and *effusion*) and function words or syntactic fillers with similar importance during patch-token alignment (Fig. 1a). Because the latter provide limited information for specific visual correspondences, these methods may introduce noisy supervision and weaken patch-token correspondence.

Recent work has attempted to differentiate token importance using corpus-level statistics. SuperCLIP [18]

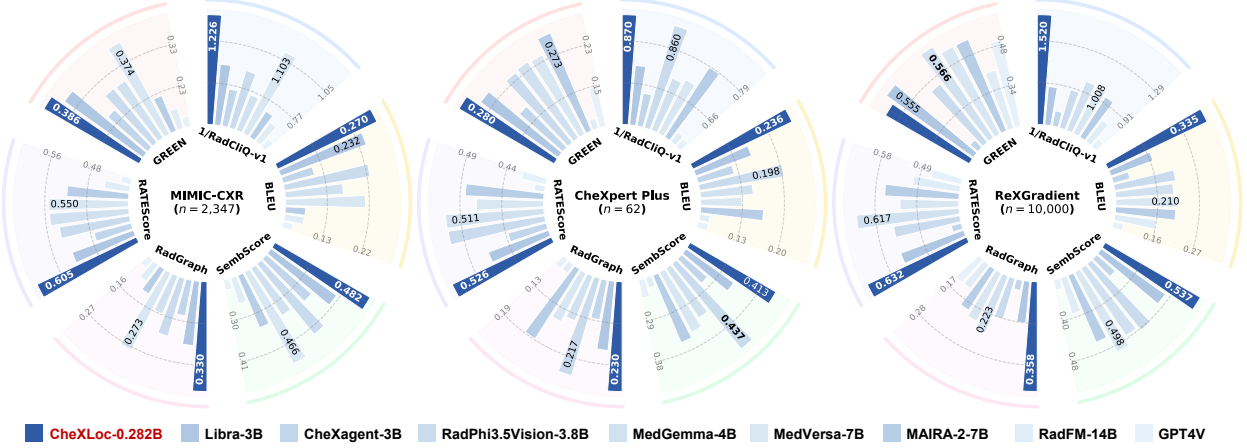
(a) Downstream task adaptation



(b) Inference efficiency for report generation

Method	Peak GPU memory (GB) ↓	Average inference time (s/sample) ↓
CheXLoc (Ours)	4.789	1.109
CheXagent-3B	5.863	1.484
MedGemma-4B	8.234	5.852
MedVersa-7B	17.528	13.133
MAIRA-2-7B	27.498	4.894

(c) Dataset-specific report generation performance



(d) Aggregate performance across report generation benchmarks

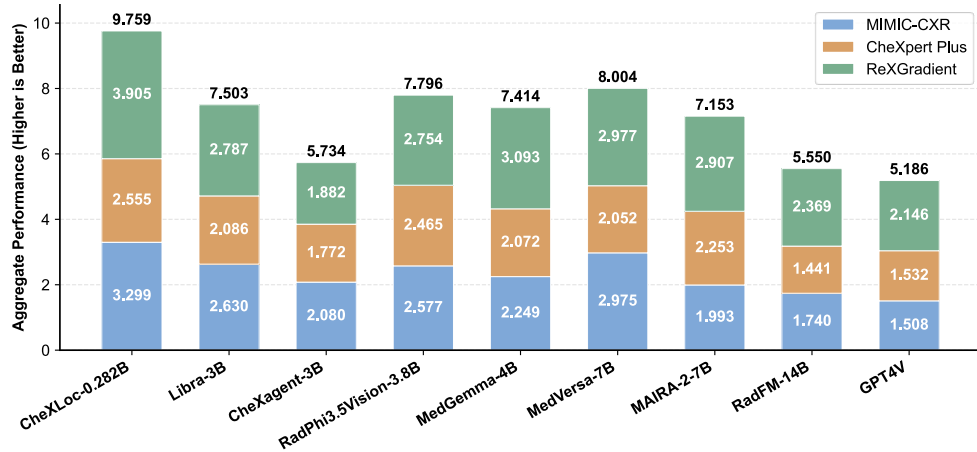


Figure 2: **Evaluation of chest X-ray report generation performance and inference efficiency.** **a**, Downstream task adaptation, in which current and, when available, prior chest X-ray images are encoded into unified visual representations and processed by task-specific prediction heads. **b**, Peak GPU memory consumption and average inference time per sample for report generation on the ReXRANK-standardized MIMIC-CXR test subset ($n = 2,347$). **c**, Quantitative comparison of report generation performance across the ReXRANK-standardized MIMIC-CXR test subset ($n = 2,347$), CheXpert Plus ($n = 62$), and ReXGradient ($n = 10,000$) using 1/RadCliQ-V1, BLEU, SemScore, RadGraph, RATEScore, and GREEN. **d**, Aggregate performance across report generation benchmarks, calculated by summing 1/RadCliQ-V1, BLEU, SemScore, RadGraph, RATEScore, and GREEN. Higher values indicate better performance.

estimates token importance using inverse document frequency. However, occurrence frequency does not directly characterize the visual supervision value of textual tokens in radiology reports. Frequently occurring terms, such as anatomical descriptions, may provide important visual cues, whereas less frequent terms, including specific findings and modifiers indicating negation or temporality, may also contribute to localized visual correspondence. Thus, frequency-based weighting alone provides an incomplete basis for estimating token importance for localized visual representation learning.

We hypothesized that clinical entities extracted from radiology reports could serve as weak supervision for learning localized visual representations by explicitly modeling their relative clinical importance during image-report alignment. To test this hypothesis, we developed CheXLoc, a report-supervised framework for learning localized visual representations from paired CXR images and radiology reports without explicit spatial annotations. CheXLoc treats clinical entities in radiology reports as weak supervisory signals and jointly optimizes global image-report alignment and entity-guided local patch-token alignment under a multiple-instance learning formulation [19, 20]. The global objective learns holistic cross-modal correspondence, whereas the local objective uses entity-guided token importance estimation to distinguish the relative importance of textual tokens, with these importance estimates determining their contributions to patch-token alignment for learning localized visual representations. This formulation encourages localized visual representations to emerge from naturally paired radiographs and reports without requiring bounding boxes, segmentation masks, or gaze annotations.

We evaluated CheXLoc across complementary clinical tasks to examine three aspects of report-supervised representation learning. First, we asked whether report-supervised pretraining supports chest X-ray report generation across internal and external cohorts, including zero-shot transfer across institutions. Second, we examined whether the learned representations transfer beyond report generation to downstream visual understanding tasks, including disease classification and lesion segmentation under limited supervision. Third, we investigated whether these representations retain visual and disease-specific evidence by assessing their sensitivity to image-removal and lesion-level perturbations. Together, these experiments test whether report supervision can support the learning of localized visual representations that generalize across datasets and institutions, transfer beyond the report-generation objective, and remain sensitive to disease-specific visual evidence.

2 Results

2.1 Report-supervised pretraining supports efficient and transferable chest X-ray report generation

We investigated whether the visual representations learned through report-supervised pretraining can support effective, efficient, and generalizable chest X-ray report generation. We evaluated CheXLoc across complementary settings, including in-domain generation on MIMIC-CXR, cross-dataset evaluation on external benchmarks, zero-shot transfer to an independent institution, and performance across heterogeneous demographic and clinical subgroups. This evaluation framework assessed both report generation quality and the transferability of the learned visual representations.

Efficient report generation on in-domain chest radiographs. We first evaluated CheXLoc on the ReXRANK-standardized MIMIC-CXR test subset [21] ($n = 2,347$), enabling direct comparison with the publicly reported benchmark results. As shown in Fig. 2c, CheXLoc achieved the highest performance across all evaluated metrics, including 1/RadCliQ-V1 [22], BLEU [23], SemScore [24], RadGraph [25], RATEScore [26], and GREEN [27]. The advantage was particularly pronounced for clinically oriented measures: CheXLoc achieved a 1/RadCliQ-V1 score of 1.226, compared with 0.898 for Libra-3B [28], 0.694 for MAIRA-2-7B [29], and 0.650 for RadFM-14B [30]. The improvements in RadGraph and GREEN indicate better agreement in extracted clinical content and fewer clinically significant report-generation errors, respectively. Importantly, these gains were achieved with only 0.282 billion parameters, substantially fewer than those of the compared multimodal foundation models.

We further evaluated CheXLoc against medical vision-language models [5, 31, 32, 12, 16] on the full MIMIC-CXR test set (Table 1; $n = 3,852$). CheXLoc achieved the highest scores across all evaluated

Table 1: **Evaluation of chest X-ray report generation performance across medical vision–language models.** Models are evaluated on the full MIMIC-CXR test set ($n = 3,852$) using five metrics. Results are reported as means with 95% confidence intervals estimated from 1,000 bootstrap samples. **Bold** denotes the **best-performing** method for each metric. Higher scores indicate better performance for all metrics.

Method	RadGraph	SembScore	1/RadCliQ-V1	RATEScore	BLEU
AFLoc	0.205 [0.201, 0.209]	0.383 [0.376, 0.390]	0.830 [0.820, 0.840]	0.534 [0.531, 0.538]	0.216 [0.212, 0.220]
RADDINO	0.197 [0.192, 0.201]	0.380 [0.373, 0.386]	0.824 [0.812, 0.836]	0.528 [0.525, 0.532]	0.225 [0.221, 0.228]
CheXWorld	0.184 [0.179, 0.187]	0.378 [0.371, 0.385]	0.805 [0.796, 0.813]	0.526 [0.522, 0.530]	0.217 [0.213, 0.220]
BiomedCLIP	0.190 [0.186, 0.195]	0.370 [0.363, 0.378]	0.796 [0.786, 0.805]	0.521 [0.518, 0.525]	0.210 [0.206, 0.213]
EGMA	0.202 [0.198, 0.205]	0.389 [0.383, 0.395]	0.763 [0.756, 0.769]	0.509 [0.506, 0.512]	0.171 [0.169, 0.174]
CheXLoc	0.329 [0.324, 0.334]	0.482 [0.475, 0.490]	1.303 [1.269, 1.327]	0.604 [0.600, 0.607]	0.340 [0.335, 0.345]

metrics (all nominal $p < 0.001$). Its RadGraph score reached 0.329, compared with 0.205 for the highest-scoring competing model, while SembScore, 1/RadCliQ-V1, and RATEScore reached 0.482, 1.303, and 0.604, respectively. The BLEU score also reached 0.340, consistent with improvements in both clinical semantic similarity and lexical correspondence with the reference reports.

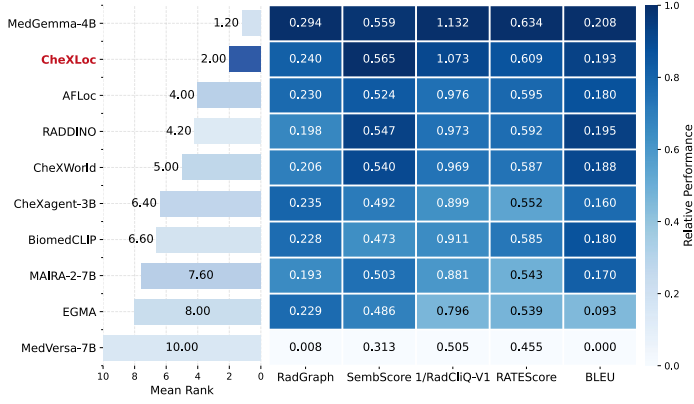
Beyond report-generation quality, CheXLoc also had substantially fewer parameters and lower GPU memory requirements than the larger multimodal models evaluated (Fig. 2b). Under the inference-time evaluation protocol described in Section 4.3, CheXLoc required an average of 1.109 s per sample, compared with 4.894 s for MAIRA-2-7B [29]. Together, these results suggest that CheXLoc achieves competitive report-generation performance while requiring substantially fewer parameters, lower GPU memory, and less inference time.

Generalization across external datasets. We next examined whether the performance gains observed on MIMIC-CXR were preserved under dataset and institutional shifts. We evaluated CheXLoc on the external CheXpert Plus [33] and ReXGradient [34] benchmarks without modifying the model architecture (Fig. 2c). On CheXpert Plus, CheXLoc achieved the highest 1/RadCliQ-V1 score (0.870) and remained competitive across the other evaluation metrics, outperforming models spanning general-purpose multimodal large language models (GPT-4V [35]), medical multimodal large language models (MedGemma-4B [36], MedVersa-7B [37], and RadFM-14B [30]), and radiology-specific report generation methods (Libra-3B [28], CheXagent-3B [38], RadPhi3.5Vision-3.8B [39], and MAIRA-2-7B [29]). On ReXGradient, a large-scale multi-institutional benchmark comprising 10,000 test studies, CheXLoc achieved the highest reported scores across all evaluation metrics, including 1/RadCliQ-V1 (1.520), BLEU (0.335), SembScore (0.537), RadGraph (0.358), and RATEScore (0.632). The consistent performance across CheXpert Plus and ReXGradient indicates that the gains observed on MIMIC-CXR extend beyond the source dataset, supporting the transferability of report-supervised visual representations across datasets and institutions.

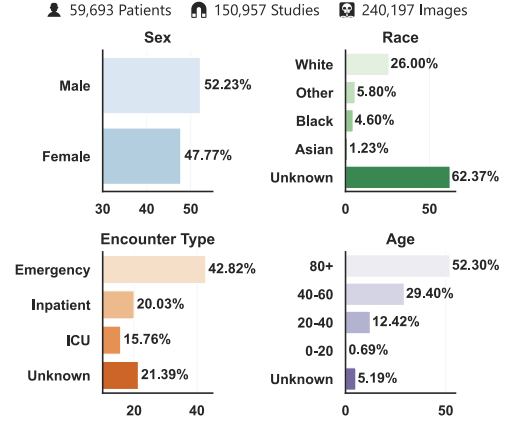
Zero-shot transfer across institutions. To further assess the transferability, we evaluated CheXLoc on the IU X-ray dataset [40] without fine-tuning (Fig. 3a; $n = 2,955$). CheXLoc ranked second among the evaluated models based on mean rank, surpassed only by MedGemma-4B [36], while requiring approximately 7% of the parameters of MedGemma-4B. These results indicate that CheXLoc retains competitive report-generation performance when transferred to an independent institutional dataset without target-domain adaptation.

Report generation performance across demographic and clinical subgroups. We next assessed whether report generation performance was consistent across heterogeneous patient populations. The demographic and clinical composition of the MIMIC-CXR pretraining cohort is summarized in Fig. 3b. The cohort included 52.23% male and 47.77% female patients, with diverse racial groups and clinical encounter types. Emergency encounters accounted for 42.82% of the cohort, and patients aged 80 years or older accounted for 52.30%. We evaluated CheXLoc across demographic and clinical subgroups in the full MIMIC-CXR test set

(a) Zero-shot report generation across institutions



(b) Pretraining cohort demographics and clinical characteristics



(c) Report generation performance across demographic and clinical characteristics

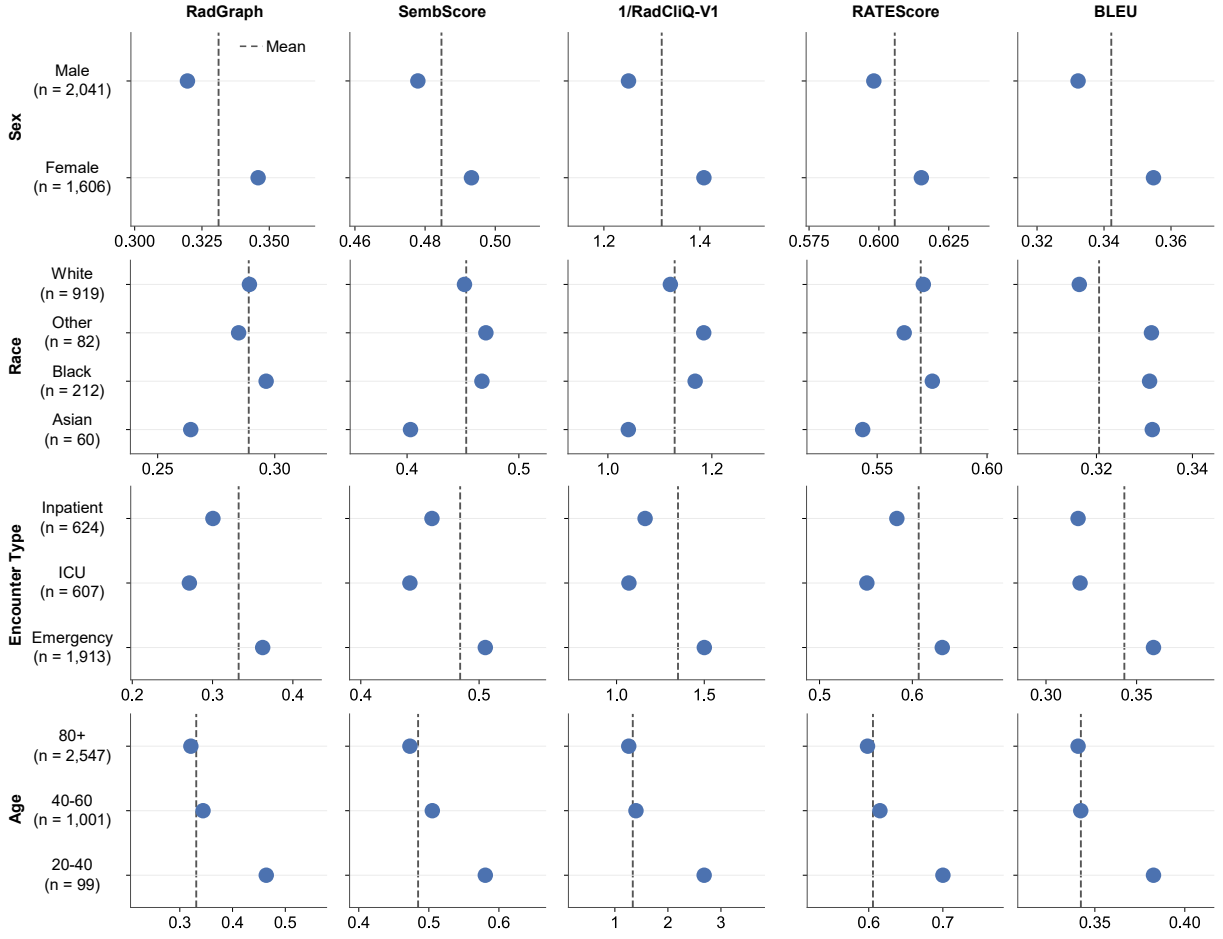
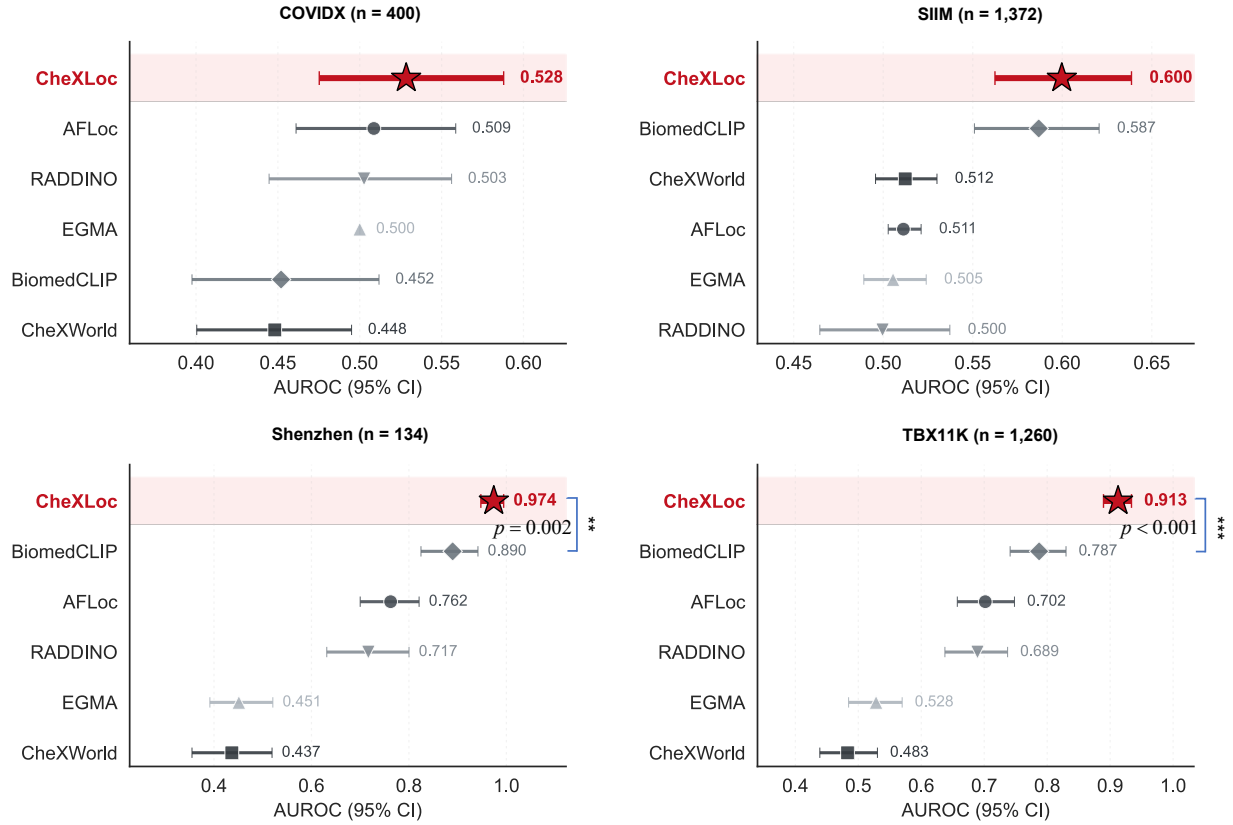


Figure 3: **Generalization of CheXLoc for report generation across institutions and patient subgroups.** **a**, Zero-shot report generation performance on the external IU X-ray dataset without additional fine-tuning ($n = 2,955$). Mean rank was computed as the average rank across all evaluation metrics. Colors indicate within-metric min-max normalized values for visualization only; numerical annotations report the original metric values. **b**, Demographics and clinical characteristics of the MIMIC-CXR training set used for pretraining. **c**, Report generation performance across demographic and clinical subgroups in the full MIMIC-CXR test set ($n = 3,852$).

(a) Zero-shot classification



(b) Supervised classification under constrained training data

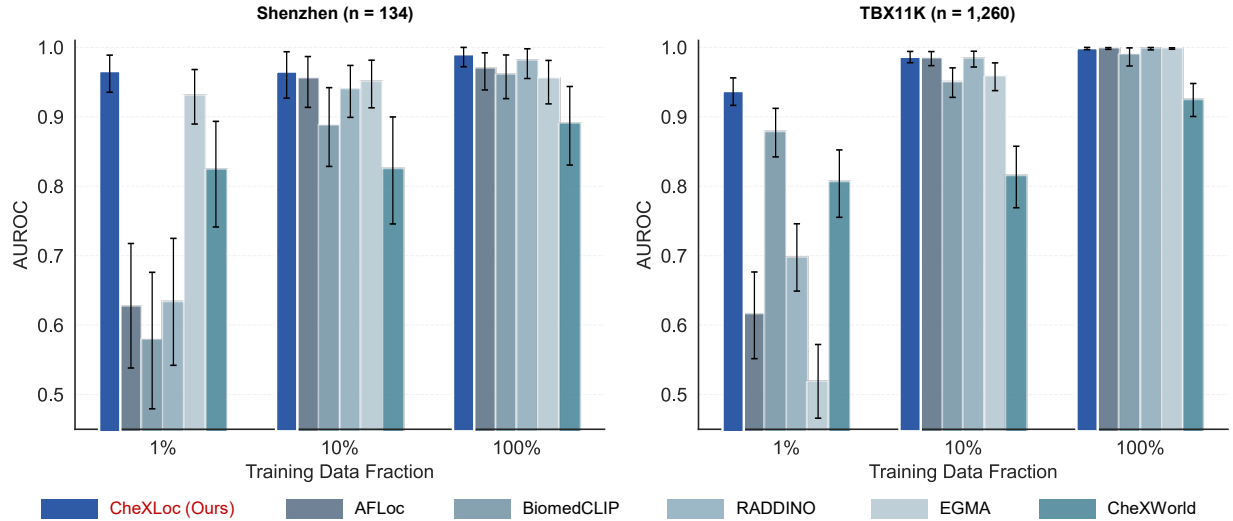


Figure 4: **Zero-shot and supervised evaluation of radiological representations across diagnostic benchmarks.** **a**, Zero-shot classification performance on four independent validation cohorts (COVIDx [41], SIIM [42], Shenzhen [43], and TBX11K [44]) for COVID-19, pneumothorax, and tuberculosis classification. **b**, Supervised classification performance on Shenzhen and TBX11K using 1%, 10%, and 100% of the downstream training data. Data points and error bars denote mean performance and 95% confidence intervals (CI) estimated from 1,000 bootstrap replicates. Statistical significance was assessed using two-sided bootstrap hypothesis testing (** : $p < 0.01$; *** : $p < 0.001$).

($n = 3,852$; Fig. 3c). No clear systematic difference was observed across the recorded subgroups, although estimates for smaller subgroups were less precise. The 1/RadCliQ-V1 scores were 1.408 for female and 1.251 for male patients, while scores across the major racial groups ranged from 1.040 to 1.184. Across encounter types and age groups, scores ranged from 1.070 for ICU encounters to 1.500 for emergency encounters, and from 1.262 among patients aged 80 years or older to 1.398 for those aged 40–60 years. The relatively high 1/RadCliQ-V1 value in the 20–40-year subgroup (2.684) should be interpreted cautiously because this subgroup contained only 99 patients. Overall, although performance varied across subgroups, no consistent pattern of lower performance was observed among the major demographic or clinical subgroups evaluated.

Taken together, these results suggest that report-supervised pretraining supports efficient and transferable chest X-ray report generation, with CheXLoc retaining strong performance across external datasets, zero-shot institutional transfer, and heterogeneous patient subgroups.

2.2 Learned visual representations transfer across diagnostic tasks

Zero-shot disease classification without disease-specific supervision. As illustrated in Fig. 4a, we first assessed whether the learned visual representations supported diagnostic classification without task-specific adaptation. CheXLoc achieved the highest AUROC across all four benchmarks, with the largest gains observed on the tuberculosis classification. On Shenzhen, CheXLoc achieved an AUROC of 0.974 (95% CI, 0.948–0.994), an 8.4-percentage-point improvement over the strongest baseline, BiomedCLIP [12] (0.890). On TBX11K, CheXLoc reached an AUROC of 0.913 (95% CI, 0.889–0.934), outperforming the strongest competing method by 12.6 percentage points. CheXLoc also achieved the highest AUROC on SIIM (0.600) and COVIDx (0.528). These results indicate that the learned visual representations support zero-shot transfer to disease classification across datasets and institutions without disease-specific supervision.

Supervised transfer under limited labelled data. We next evaluated the transferability of the pre-trained representations under varying amounts of labelled data (Fig. 4b). All models were fine-tuned on the Shenzhen [43] and TBX11K [44] benchmarks using 1%, 10%, or 100% of the downstream training data. With 1% of the training data, CheXLoc achieved AUROCs of 0.966 (95% CI, 0.935–0.989) on Shenzhen and 0.937 (95% CI, 0.917–0.956) on TBX11K, exceeding the strongest comparator by 3.5 and 5.8 percentage points, respectively.

As the amount of labelled data increased, performance differences between methods narrowed. Under 10% supervision, several medical vision–language models approached CheXLoc, particularly on TBX11K. With full supervision, leading methods reached near-ceiling performance on TBX11K (AUROC ≥ 0.99), whereas CheXLoc achieved the highest AUROC on Shenzhen (0.990). Thus, CheXLoc showed its largest advantage under limited supervision, with performance differences diminishing as more labelled data became available.

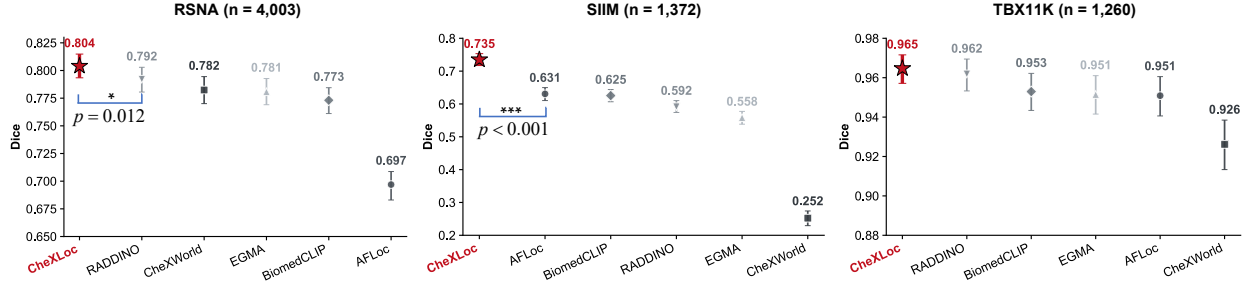
2.3 Lesion segmentation across thoracic diseases

CheXLoc achieved the highest mean Dice score across the RSNA [45], SIIM [42], and TBX11K [44] benchmarks (Fig. 5a). On RSNA, CheXLoc achieved a Dice score of 0.804 (95% CI, 0.793–0.815), compared with 0.792 for RADDINO [31] ($p = 0.012$). On SIIM, CheXLoc achieved 0.735, 10.4 percentage points higher than the strongest baseline ($p < 0.001$). On TBX11K, CheXLoc achieved 0.965, compared with 0.962 for RADDINO. Qualitative results were consistent with these findings, with CheXLoc showing more complete lesion coverage and fewer apparent false-negative regions across the three benchmarks (Fig. 5b). These results suggest that learned visual representations transfer to lesion segmentation across different thoracic diseases and annotation formats.

2.4 Sensitivity of report generation to visual and disease-specific evidence

We evaluated disease-specific visual evidence at two complementary levels. We first quantified the dependence of report generation on visual input by removing the radiograph and then examined whether generated findings responded to disease-specific pathological regions using lesion-level interventions.

(a) Qualitative lesion segmentation results



(b) Quantitative evaluation of lesion segmentation

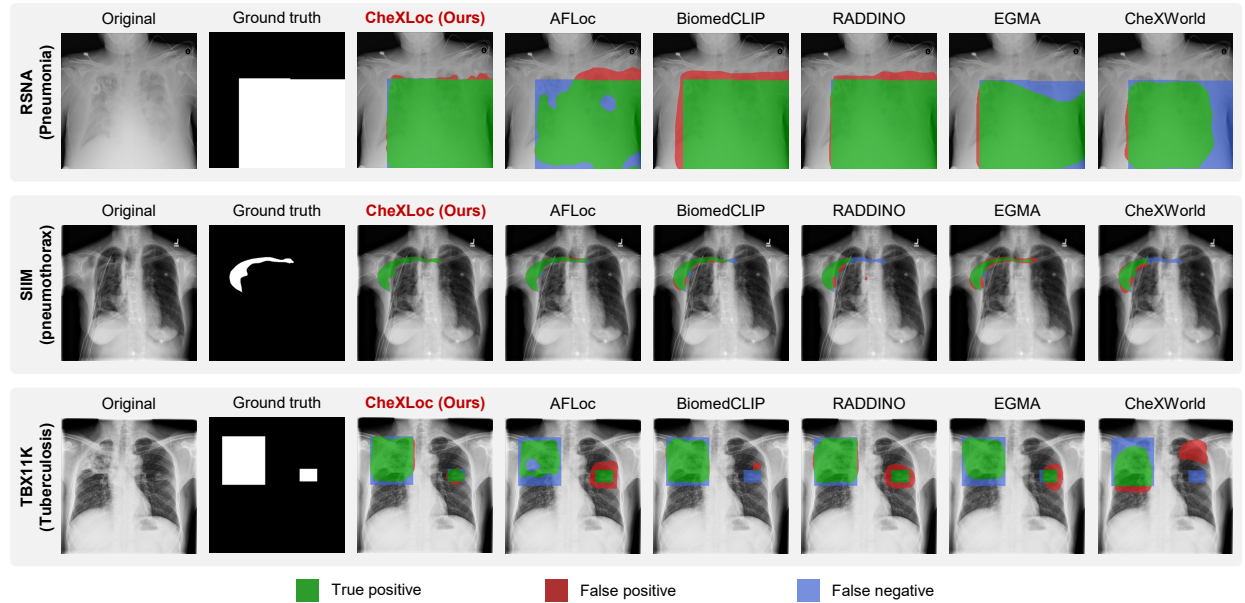


Figure 5: **Lesion segmentation across diverse thoracic diseases.** Pixel-level or box-derived lesion annotations were used exclusively for downstream supervised segmentation and were not used during CheXLoc pre-training. **a**, Quantitative comparison of lesion segmentation performance on the RSNA, SIIM, and TBX11K benchmarks. Points denote the mean Dice score, and error bars indicate the 95% confidence interval estimated from 1,000 bootstrap replicates. **b**, Representative lesion segmentation results for CheXLoc and existing medical vision–language models. Reference annotations are shown as box-derived masks for RSNA and TBX11K and pixel-level masks for SIIM.

2.4.1 Sensitivity of report generation to visual input

Quantitative analysis. To quantify the contribution of visual input to report generation, we evaluated CheXLoc on the ReXRank-standardized MIMIC-CXR test subset ($n = 2,347$) using the original radiograph and a blank image, while keeping all other generation settings constant. We quantified visual dependence as the difference in report quality between the original and blank-image conditions:

$$\Delta(1/\text{RadCliQ-V1}) = (1/\text{RadCliQ-V1})_{\text{Original}} - (1/\text{RadCliQ-V1})_{\text{Blank}}. \quad (1)$$

Larger positive values indicate a greater sensitivity of report quality to the removal of visual input. As shown in Fig. 6a, CheXLoc showed the largest difference between the original and blank-image conditions ($\Delta = 0.434$), compared with 0.019–0.235 for the other models, indicating a stronger dependence of report quality on visual input.

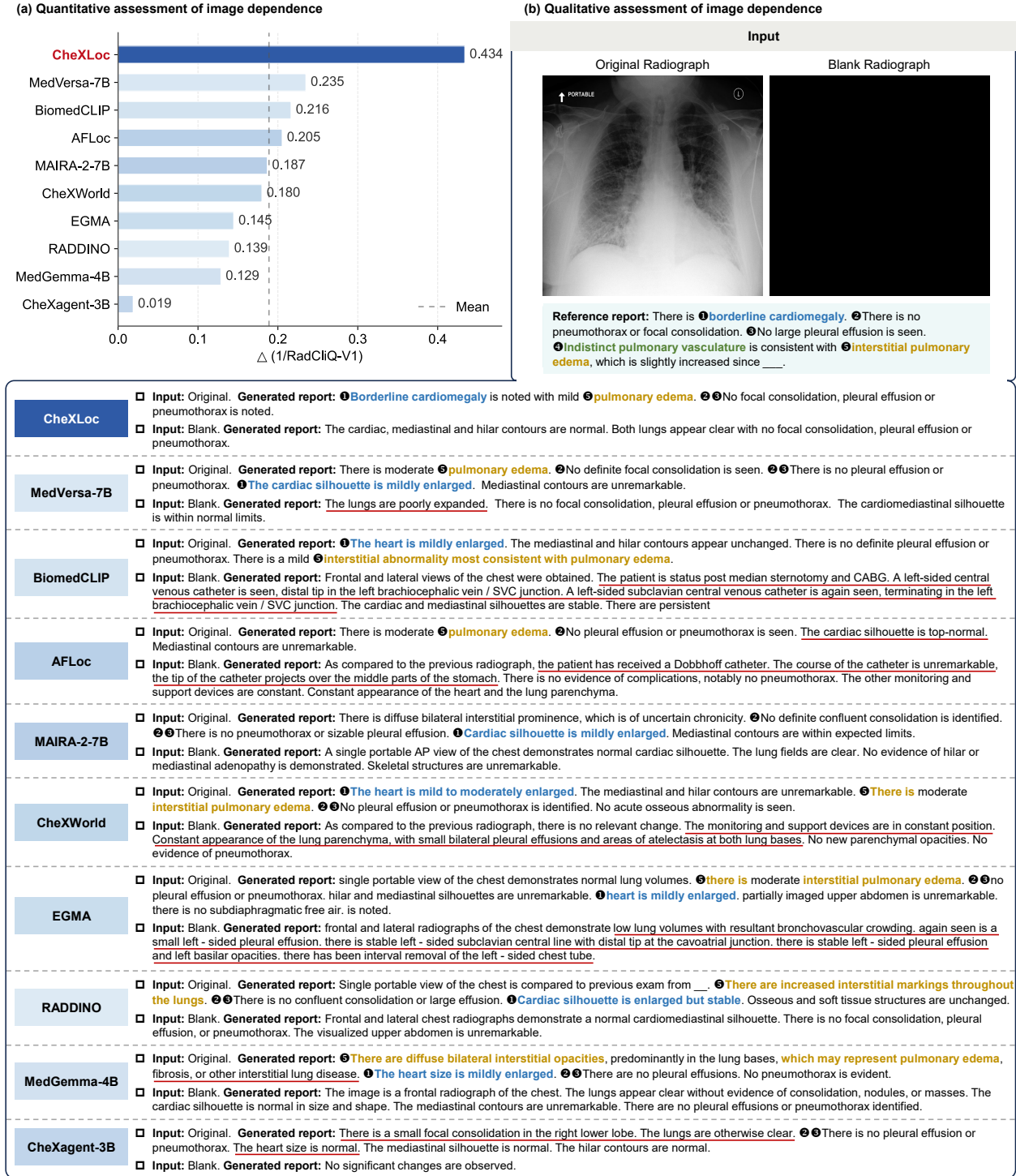


Figure 6: Assessing image dependence in report generation. a, Quantitative assessment of visual dependence using $\Delta(1/\text{RadCliQ-V1})$ between reports generated from original and blank radiographs on the ReXRank-standardized MIMIC-CXR test subset ($n = 2,347$). Larger positive values indicate a greater reliance on visual information. b, Qualitative comparison of the corresponding generated reports. Numbered sentences indicate correct correspondence with sentences in the reference report, while coloured text highlights correctly generated abnormal findings or devices. Clinically significant errors are marked with red underlining.

Qualitative analysis. We next examined model responses to the presence or absence of visual input by comparing reports generated from the original and blank radiographs (Fig. 6b). With the original radiograph, CheXLoc captured the borderline cardiomegaly and edema described in the reference report without overstating its severity. Several baseline models also identified these abnormalities but tended to overstate cardiomegaly, for example, by describing borderline cardiomegaly as mild cardiomegaly. RADDINO [31] additionally generated *stable cardiomegaly*, which was not presented in the reference report, whereas AFLoc [5], MAIRA-2-7B [29], and CheXagent-3B [38] omitted one or more clinically relevant findings.

With the blank radiograph, CheXLoc, MAIRA-2-7B, RADDINO, MedGemma-4B, and CheXagent-3B generated reports without introducing abnormal findings. Ideally, a blank radiograph would prompt the model to recognize the invalid input and request a valid chest radiograph; however, such behaviour was not explicitly supervised during training. Because a blank radiograph provides no visual evidence of abnormal findings, generating normal findings can still be a reasonable response in this setting. In contrast, MedVersa-7B, BiomedCLIP, AFLoc, CheXWorld, and EGMA introduced abnormal findings despite the absence of corresponding visual evidence. These qualitative observations are consistent with the quantitative finding that CheXLoc showed the largest difference in report quality between the original and blank-image conditions.

2.4.2 Disease-specific localization awareness in report generation

Quantitative analysis. To assess whether report generation was sensitive to disease-specific visual regions, we evaluated nine medical foundation models on the MS-CXR benchmark [46] using the Localization-Aware Generation Evaluation (LAGE; Sec. 4.6). LAGE quantifies the response of generated findings to lesion-level perturbations using complementary lesion-only and lesion-removed interventions.

As shown in Fig. 7a, CheXLoc achieved the highest LAGE score for pneumothorax (0.377), with Sufficiency and Necessity scores of 0.461 and 0.294, respectively. Most baseline models had LAGE scores below 0.200. For pleural effusion, lung opacity, and edema, CheXLoc achieved the second- or third-highest LAGE score, with differences of 0.023, 0.041, and 0.010, respectively, from the highest-scoring model, MAIRA-2-7B [29]. Notably, MAIRA-2-7B, MedVersa-7B [37], and EGMA [16] incorporate spatial supervision during pretraining, whereas CheXLoc relies solely on image-report pairs. Despite this difference in supervision, CheXLoc achieved comparable LAGE scores for the evaluated findings, supporting the use of report supervision as a weak signal for learning representations sensitive to disease-specific visual regions.

Qualitative analysis. We further examined localization-aware report generation using lesion-only and lesion-removed perturbations for representative cases of pneumothorax and edema (Fig. 7b). For pneumothorax (Case 1), CheXLoc and MAIRA-2-7B [29] generated *left apical pneumothorax* from the lesion-only image, whereas EGMA [16] failed to recover the target finding. After lesion removal, CheXLoc suppressed the pneumothorax finding while preserving the remaining reported abnormalities. In contrast, MAIRA-2-7B continued to generate the spurious finding of pneumothorax after lesion removal, whereas EGMA omitted *left pleural effusion*.

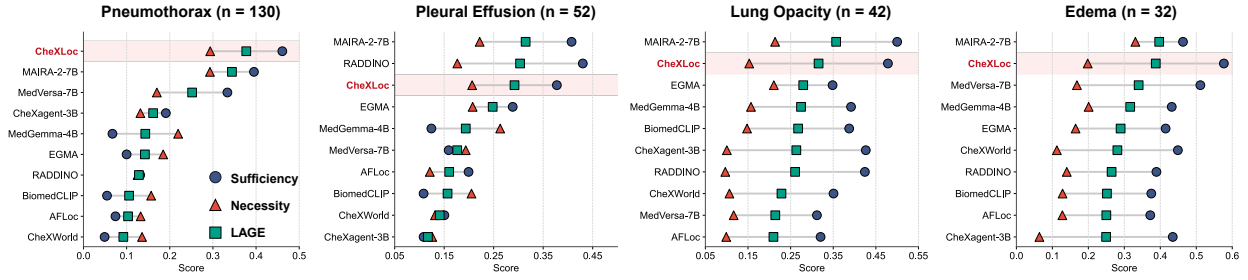
For edema (Case 2), all three models recovered the edema-related finding from the lesion-only image. Because the edema lesion covered a relatively extensive region spanning the thorax, lungs, and abdomen, the lesion-only image retained visual information beyond the target abnormality. CheXLoc therefore also correctly recovered other findings, including *catheter tip terminates in the right atrium*, *aortic unfolding with calcification at the aortic knob*, and *small pleural effusion*, whereas MAIRA-2-7B and EGMA failed to recover these findings. After lesion removal, all three models suppressed the edema-related finding. These examples suggest that CheXLoc can respond to the targeted abnormality while retaining other abnormalities supported by the available visual information.

Together, the quantitative and qualitative results support the sensitivity of CheXLoc’s generated findings to disease-specific visual regions, particularly for focal abnormalities such as pneumothorax, while retaining sensitivity to more diffuse findings such as edema.

2.5 Ablation studies

We conducted controlled ablation experiments to characterize the contributions of the two alignment objectives in CheXLoc and to examine the design choices underlying entity-guided local patch-token alignment.

(a) Disease-specific localization awareness in report generation



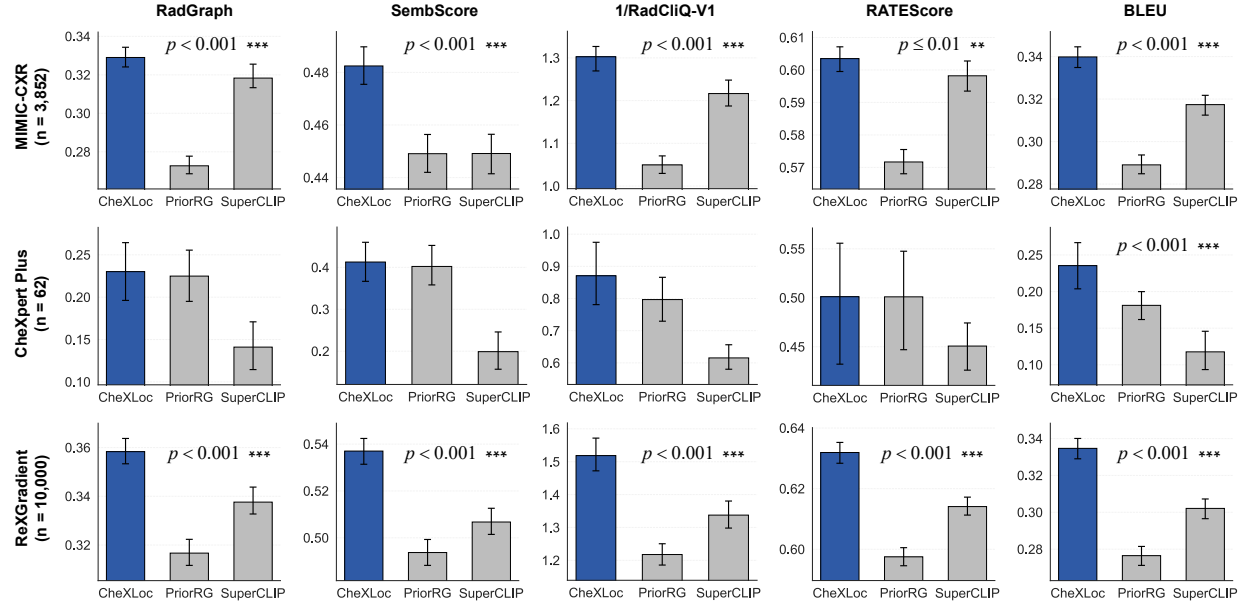
(b) Quantitative evaluation of disease-specific localization awareness



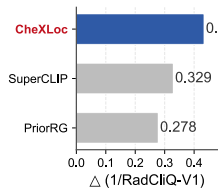
Figure 7: Disease-specific localization awareness in chest X-ray report generation. a, Localization awareness of nine medical foundation models across four thoracic diseases, evaluated using Sufficiency, Necessity, and the integrated LAGE score. b, Qualitative examples of localization-aware report generation under lesion-only and lesion-removed interventions. Numbered sentences indicate correct correspondence with sentences in the reference report, while coloured text highlights correctly generated abnormal findings or devices. Clinically significant errors are marked with red underlining.

Specifically, we investigated: (i) whether entity-guided local patch–token alignment provides benefits beyond global image–report alignment; (ii) whether explicitly incorporating clinical entities into token importance estimation provides advantages over frequency-based token weighting; (iii) the individual and complementary

(a) Report generation performance



(b) Visual dependence



(c) Disease-specific localization awareness in report generation

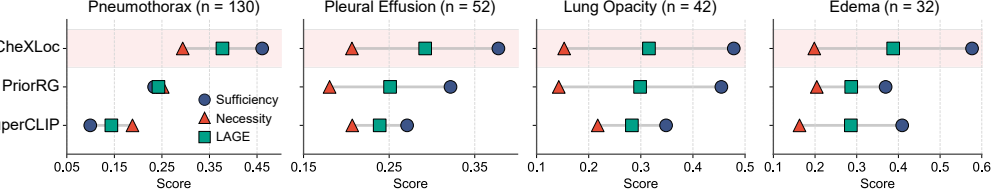


Figure 8: **Comparison of global and token-weighted visual–textual alignment strategies.** CheXLoc is compared with PriorRG [13], representing a global alignment baseline, and SuperCLIP [18], a frequency-based token-weighting baseline. These evaluations span report generation quality, visual dependence, and disease-specific regional sensitivity. **a**, Report generation on internal and external datasets; **b**, Visual dependence quantified by $\Delta(1/RadCliQ-V1)$ between original and blank radiographs ($n = 2,347$), with larger values indicating greater reliance on visual information; **c**, Disease-specific sensitivity to localized evidence across four thoracic diseases, assessed using Sufficiency, Necessity, and LAGE.

contributions of global image–report alignment and entity-guided local patch–token alignment; and (iv) the design choices underlying the entity-guided local patch–token alignment objective.

2.5.1 Effect of entity-guided local patch–token alignment

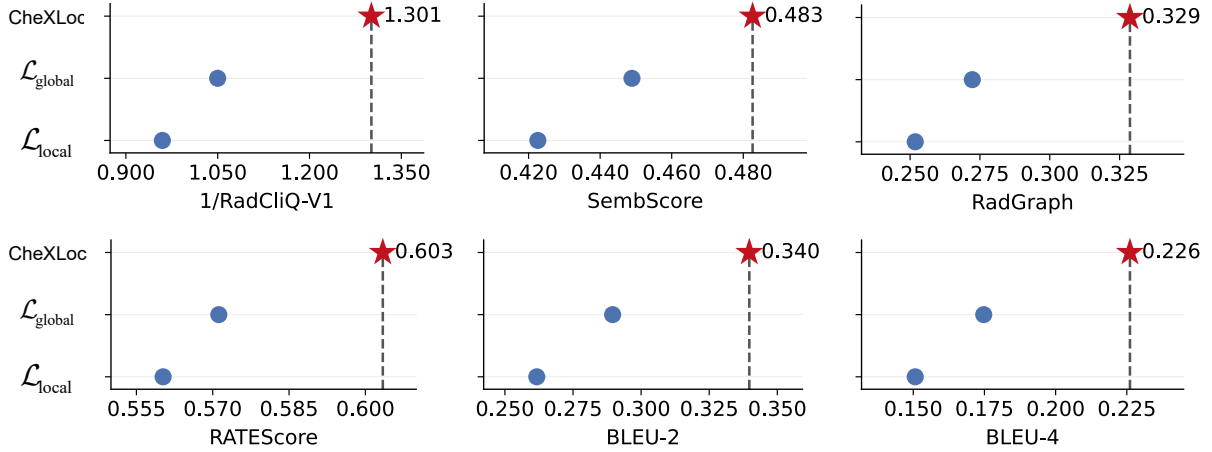
We compared CheXLoc with PriorRG [13] across three complementary dimensions: in-domain and external report generation, visual dependence, and disease-specific visual localization (Fig. 8). PriorRG shares the same visual encoder, text encoder, and report generator as CheXLoc, but performs cross-modal alignment exclusively at the global level. CheXLoc extends this formulation with entity-guided local patch–token alignment, which establishes fine-grained correspondence between clinically meaningful textual tokens and localized visual representations.

For report generation, CheXLoc consistently outperformed PriorRG on the full MIMIC-CXR test set ($n = 3,852$) and the external ReXGradient dataset ($n = 10,000$), with statistically significant improvements across the evaluated lexical and clinically oriented metrics ($p < 0.001$). In particular, $1/RadCliQ-V1$ increased by 25.3% on MIMIC-CXR and 30.1% on ReXGradient. On the smaller external CheXpert Plus cohort ($n = 62$), CheXLoc achieved the highest overall performance, although the difference from PriorRG did not

reach statistical significance.

The improvement in report generation was accompanied by stronger visual correspondence. CheXLoc exhibited substantially greater visual dependence than PriorRG (0.434 versus 0.278) and achieved higher LAGE scores for all four evaluated diseases, including edema (0.387 versus 0.287), lung opacity (0.316 versus 0.299), pleural effusion (0.292 versus 0.251), and pneumothorax (0.377 versus 0.243). Together, these results show that incorporating entity-guided local patch-token alignment into global image-report alignment improves report-generation metrics and increases sensitivity to visual evidence.

(a) Contributions of global image-report alignment and entity-guided local patch-token alignment



(b) Disentangling entity-guided local patch-token alignment

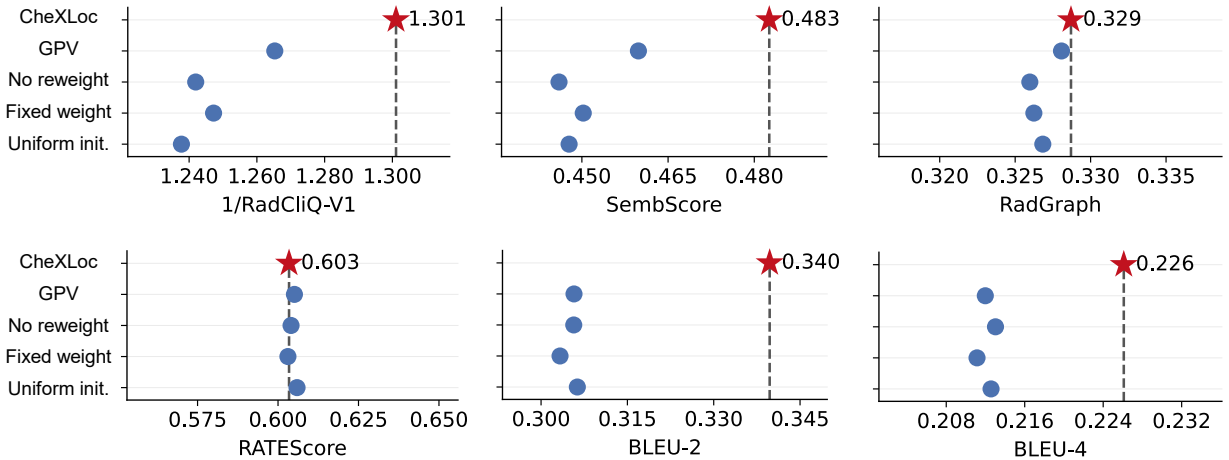


Figure 9: **Ablation analysis of global image-report alignment ($\mathcal{L}_{\text{global}}$) and entity-guided local patch-token alignment ($\mathcal{L}_{\text{local}}$) on the full MIMIC-CXR test set ($n = 3,852$).** **a**, Comparison of $\mathcal{L}_{\text{global}}$, $\mathcal{L}_{\text{local}}$, and CheXLoc ($\mathcal{L}_{\text{global}} + \mathcal{L}_{\text{local}}$); **b**, Disentangling entity-guided local patch-token alignment. GPV denotes globally pooled visual features used for probability prediction. Uniform init. denotes $\alpha(\cdot)$ initialized to one across all entity types and optimized during training.

2.5.2 Influence of clinical entity guidance in token importance estimation

We compared CheXLoc with SuperCLIP [18] (Fig. 8). SuperCLIP estimates token importance from frequency-based statistics, whereas CheXLoc incorporates clinical entities to estimate token importance from visual-textual correspondence.

On the full MIMIC-CXR test set ($n = 3,852$), CheXLoc significantly outperformed SuperCLIP across the

evaluated lexical and clinically oriented metrics ($p \leq 0.01$), with RadGraph increasing from 0.318 to 0.329, 1/RadCliQ-V1 from 1.216 to 1.303, and SembScore from 0.449 to 0.482. Similar improvements were observed on the external CheXpert Plus and ReXGradient datasets, where CheXLoc consistently outperformed SuperCLIP across lexical and clinically oriented metrics ($p < 0.001$).

The advantage of clinical entity guidance was also reflected in visual evidence. CheXLoc exhibited greater visual dependence than SuperCLIP (0.434 versus 0.329) and achieved higher LAGE scores across all four evaluated diseases. Together, these results suggest that explicitly incorporating clinical entities into token importance estimation improves both report-generation performance and sensitivity to disease-specific visual evidence compared with frequency-based token weighting in the evaluated settings.

2.5.3 Contributions of global image-report alignment and entity-guided local patch-token alignment

To dissect the contributions of the two alignment objectives, we compared CheXLoc with two variants trained using either $\mathcal{L}_{\text{global}}$ or $\mathcal{L}_{\text{local}}$ alone on the full MIMIC-CXR test set ($n = 3,852$; Fig. 9). The global-only variant achieved RadGraph, SembScore, 1/RadCliQ-V1, RATEScore, BLEU-2, and BLEU-4 scores of 0.272, 0.449, 1.050, 0.571, 0.290 and 0.175, respectively, whereas the local-only variant achieved 0.252, 0.423, 0.960, 0.560, 0.262 and 0.151, respectively. In contrast, jointly $\mathcal{L}_{\text{global}}$ and $\mathcal{L}_{\text{local}}$ consistently yielded the highest scores across all six metrics, reaching 0.329, 0.483, 1.301, 0.603, 0.340, and 0.226, respectively. Together, these results support complementary roles for the two alignment objectives: global image-report alignment establishes holistic cross-modal correspondence, whereas entity-guided local patch-token alignment further refines the correspondence between localized visual representations and clinical entities.

2.5.4 Disentangling entity-guided local patch-token alignment

We next dissected the design choices underlying entity-guided local patch-token alignment (Fig. 9b). We first examined the contribution of spatially localized visual features by replacing patch-level features with globally pooled visual features (GPV). This modification consistently reduced performance on the evaluated metrics, with SembScore decreasing from 0.483 to 0.460, 1/RadCliQ-V1 from 1.301 to 1.265, and BLEU-4 from 0.226 to 0.212. These results suggest that patch-level features provide more informative visual evidence for report generation.

We next investigated the role of token reweighting in differentiating the relative importance of clinically relevant tokens. We removed the reweighting operation while retaining only the binary cross-entropy loss in Eq. (9), yielding the No reweight variant. This variant identifies whether a token is clinically relevant but does not distinguish its relative importance. Removing the reweighting operation consistently degraded performance, with SembScore decreasing from 0.483 to 0.446, 1/RadCliQ-V1 from 1.301 to 1.242, and BLEU-2 from 0.340 to 0.306. These results suggest that identifying clinically relevant tokens alone is insufficient and that modeling their relative importance contributes to improved report generation performance.

We next examined whether expert-informed initialization provides a useful prior for learning entity importance. We initialized the importance mapping $\alpha(\cdot)$ to one for all entity types, yielding the Uniform init variant. This variant reduced SembScore from 0.483 to 0.448, 1/RadCliQ-V1 from 1.301 to 1.238, and BLEU-4 from 0.226 to 0.213. Despite allowing the importance mapping to be optimized throughout training, Uniform init consistently underperformed CheXLoc, suggesting that expert-informed initialization provides a more informative starting point for learning entity importance than uniform initialization.

Finally, we assessed whether allowing the expert-informed importance mapping $\alpha(\cdot)$ to adapt during training improves report generation. We initialized $\alpha(\cdot)$ with the expert-informed weights but kept it fixed throughout training, yielding the Fixed weight variant. This variant reduced SembScore from 0.483 to 0.450, 1/RadCliQ-V1 from 1.301 to 1.247, and BLEU-2 from 0.340 to 0.303. These results suggest that although the expert-informed weights provide a useful prior, allowing $\alpha(\cdot)$ to adapt during training contributes to improved report generation performance.

3 Discussion

Learning localized visual representations from radiology reports without explicit spatial supervision requires models to recover image-report correspondences from inherently indirect and heterogeneous signals. Our results suggest that clinical entities within radiology reports can provide such weak supervision. CheXLoc therefore uses entity-guided token importance estimation to distinguish the relative informativeness of textual entities and modulate their contributions to local patch-token alignment, while global image-report alignment preserves holistic cross-modal correspondence. The resulting representations transfer across report generation, disease classification, and lesion segmentation, supporting report supervision as a source of weak supervision for localized visual representation learning without explicit spatial annotations. Here, localized representation learning denotes patch-level visual features that are sensitive to clinically informative regions, rather than validated spatial correspondence between individual textual entities and anatomical or pathological regions.

The ablation experiments support complementary roles for global image-report alignment and entity-guided local patch-token alignment. The global objective preserves holistic image-report correspondence, whereas the local objective uses entity-guided token importance estimates to modulate the contribution of individual textual tokens during patch-token alignment, promoting localized visual representation learning. The consistent benefits observed from patch-level representations, token reweighting, expert-informed initialization, and learnable adaptation of the importance mapping further support the design of the entity-guided local patch-token alignment objective. The results suggest that effective localized representation learning may benefit not only from retaining patch-level visual representations, but also from differentiating the relative importance of textual tokens according to their clinical relevance.

The learned representations also transferred across tasks and supervision regimes. CheXLoc retained strong performance across external report-generation datasets and under zero-shot transfer to an independent institution, despite being pretrained only on MIMIC-CXR. Its representations also transferred to disease classification and lesion segmentation, with the relative advantage most apparent when downstream classification labels were scarce. As more labelled data became available, performance differences between methods narrowed. These findings suggest that report-supervised pretraining may be particularly useful when paired image-report data are more readily available than task-specific diagnostic or spatial annotations. At the same time, the transfer results should be interpreted as evidence of reusable visual representations rather than as evidence that report supervision can replace task-specific diagnostic or spatial annotation.

The perturbation experiments further support the sensitivity of CheXLoc to visual and disease-specific evidence. Replacing radiographs with blank images led to a larger reduction in report quality for CheXLoc, while lesion-level perturbations produced stronger responses for focal abnormalities such as pneumothorax. The smaller differences observed for more diffuse findings, such as edema and pleural effusion, are consistent with their broader spatial distribution, which may reduce the specificity of lesion-level perturbations. Together, these results suggest that CheXLoc is sensitive to visual evidence at both image and lesion levels, while also highlighting that perturbation-based evaluation alone cannot establish causal image-report grounding.

Several limitations should be considered. CheXLoc relies on automated clinical entity extraction and a manually constructed lexicon, and errors or omissions in these components may affect the weak supervisory signal. The current study is also limited to chest radiographs and predominantly English-language reports, and the generalizability of the approach to other imaging modalities, anatomical regions, and languages remains to be established. In addition, the lesion-level perturbation analysis uses predefined pathological regions and therefore provides evidence of localization awareness rather than direct validation of fine-grained image-report correspondence. The LAGE analysis further relies on a counterfactual assumption that removing a disease-related region selectively affects the corresponding finding while leaving other clinical findings unchanged; this assumption may not always hold because pathological findings can be spatially and clinically interrelated. Future work should evaluate the clinical utility of CheXLoc in prospective real-world workflows and investigate its applicability across broader imaging and clinical settings.

4 Methods

4.1 Architecture of CheXLoc

Overview. CheXLoc is designed to learn localized visual representations from paired chest radiographs and radiology reports without requiring spatial annotations. The central hypothesis is that textual elements in radiology reports provide heterogeneous levels of supervision for visual representation learning. In particular, clinical entities describing radiographic findings are hypothesized to provide more direct supervision for localized visual correspondence, whereas function words and syntactic fillers, such as articles and prepositions, are expected to provide limited information for establishing specific visual correspondences. Consequently, conventional image-report alignment strategies [17, 5] that assign uniform importance to all textual tokens may dilute clinically informative cross-modal correspondences and hinder the learning of localized visual representations.

CheXLoc draws on weakly supervised localization [47, 48], which enables localized visual representations to be learned from coarse-level supervision without explicit spatial annotations. To model the correspondence between clinical entities and localized visual evidence, CheXLoc further adopts a multiple-instance learning formulation [19, 20], treating a CXR as a bag of image patches and clinical entities as coarse semantic supervision. For each entity, only a subset of patches is expected to provide relevant visual evidence, without explicit entity-patch annotations. This formulation motivates learning entity-patch correspondence through cross-modal alignment. To this end, CheXLoc consists of two complementary components: (i) a global image-report alignment module that learns holistic cross-modal correspondence, and (ii) an entity-guided local patch-token alignment objective that uses entity-guided token importance estimation to distinguish the relative importance of textual tokens, with these importance estimates determining their contributions to patch-token alignment for learning localized visual representations. No manually annotated spatial labels were used during pretraining; all token-level supervision was deterministically derived from the accompanying radiology reports.

Text encoding. We employ a shared language encoder to obtain representations of radiology reports and associated clinical context. To explicitly distinguish different clinical sections, task-specific special tokens ([Findings], [Indication], [History], [Comparison], and [Technique]) are prepended to the corresponding input sequences.

The augmented textual inputs are encoded using CXR-BERT [49]. The resulting representations are projected into a shared vision-language embedding space with dimension d through multilayer perceptron (MLP) projection heads. We denote the resulting report and contextual representations as $\mathbf{T}^r \in \mathbb{R}^{N^r \times d}$ and $\mathbf{T}^c \in \mathbb{R}^{N^c \times d}$, where N^r and N^c represent the corresponding sequence lengths.

Image encoding. Following the protocol of PriorRG [13], we utilize RADDINO [31] as the visual encoder to extract patch-level representations from the current radiograph and, when available, the most recent prior examination. When prior examinations are unavailable, the model operates using the current examination alone. The extracted visual features are projected into the shared embedding space through an MLP projection head. To integrate information across serial examinations, a Vision Transformer block [50] is employed to model interactions among visual tokens from different time points. The resulting visual representation is denoted as $\mathbf{X} \in \mathbb{R}^{N^v \times d}$, where N^v represents the number of image patches.

Global image-report alignment. Radiological interpretation is inherently contextual, relying on information such as clinical symptoms, examination indications, prior history, and imaging techniques. To incorporate such heterogeneous clinical context, we adopt the context-guided multimodal fusion module introduced in PriorRG [13]. Specifically, a Perceiver module [51] with learnable latent queries aggregates the contextual representations $\mathbf{T}^c$ and subsequently fuses the resulting context representation with visual features $\mathbf{X}$ to obtain a global multimodal representation. We then optimize global image-report correspondence using a multi-positive contrastive objective [52, 13], denoted as $\mathcal{L}_{\text{global}}$, which promotes similarity between matched image-report pairs while distinguishing them from unmatched pairs within each mini-batch.

Entity-guided token importance estimation. Existing token-level alignment methods [17, 5] generally assume that all textual tokens contribute equally to image–report correspondence. However, radiology reports contain substantial semantic heterogeneity. Clinical entities describing radiographic findings (e.g., *opacity*, *effusion*, and *cardiomegaly*) provide weak semantic supervision, whereas function words and syntactic fillers contribute little to visual correspondence. Therefore, assigning equal importance to all textual tokens during alignment may introduce noisy supervision and interfere with effective patch–token correspondence.

To address this limitation, CheXLoc introduces an entity-guided token importance estimation strategy to distinguish the relative importance of textual tokens. Unlike frequency-based weighting approaches [18] such as inverse document frequency, which derive token weights from occurrence frequency, our approach leverages clinical entities from radiology reports to distinguish clinically relevant tokens, as occurrence frequency alone may not adequately reflect clinical relevance. Specifically, RadGraph [25] is used to identify clinical entities and their entity types from radiology reports. Because some clinically meaningful expressions, such as *no* or *stable*, may not be captured by named entity recognition, we further incorporate a manually constructed clinical lexicon to recover additional terms, including negation and temporal descriptors. For the i^{th} sample, the entity–type pairs extracted from the corresponding report are represented as:

$$\{(e_{i,j}, r_{i,j})\}_{j=1}^{N_i^e}, r_{i,j} \in \mathcal{T}, \quad (2)$$

where $e_{i,j}$ denotes the j^{th} extracted entity span, $r_{i,j}$ indicates its entity type, $\mathcal{T}$ denotes the set of entity types, and N_i^e represents the total number of entities. To account for the unequal contributions of different entity types to visual–textual correspondence, we define a learnable type-specific importance function $\alpha : \mathcal{T} \rightarrow \mathbb{R}_+$ that assigns an importance weight to each entity type. Accordingly, the j^{th} entity in the i^{th} sample is assigned the weight $\alpha(r_{i,j})$. The importance mapping $\alpha(\cdot)$ is optimized with the entire framework during pretraining, allowing the model to learn the relative contribution of each entity type from image–report pairs.

Because vision–language pretraining operates on individual tokens, we transfer the importance assigned to each clinical entity to the tokenizer output tokens that constitute the entity. This formulation enables the model to learn fine-grained correspondence between visual patches and clinically relevant textual units, while maintaining compatibility with autoregressive report generation. Specifically, for each vocabulary token k , we aggregate its entity-derived importance across all occurrences in the report. Formally,

$$\mathbf{w}_{i,k} = \sum_{j=1}^{N_i^e} \alpha(r_{i,j}) \cdot \text{Freq}(k, e_{i,j}), k \in \mathcal{V}, \quad (3)$$

where $\text{Freq}(k, e_{i,j})$ denotes the number of occurrences of token k within entity span $e_{i,j}$, and $\mathcal{V}$ represents the vocabulary. This mapping converts entity-level clinical supervision into token-level importance weights.

Entity-guided local patch–token alignment objective. The estimated token importance $\mathbf{w}$ determines the contribution of each textual token to patch–token alignment for learning localized visual representations. Specifically, patch-level visual representations $\mathbf{X}$ are used to predict token-level clinical semantics, with the contribution of each token adaptively modulated according to its clinical importance and prediction difficulty.

We first derive binary token-level targets from the estimated token importance scores:

$$\mathbf{b} = \mathbb{I}(\mathbf{w} > 0) \in \mathbb{R}^{B \times |\mathcal{V}|}, \quad (4)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function, B is the batch size, and $|\mathcal{V}|$ represents the vocabulary size. Thus, $\mathbf{b}_{i,j} = 1$ indicates the j^{th} token is part of a clinical entity in the i^{th} sample. To account for both clinical importance and prediction difficulty, we introduce an adaptive reweighting factor $\beta \in \mathbb{R}^{B \times |\mathcal{V}|}$. Unlike a probability distribution, β serves as an optimization weight that modulates the contribution of individual tokens to the training objective. We first transform the token importance scores as follows:

$$\tilde{\mathbf{w}} = \log(1 + \mathbf{w}), \quad (5)$$

$$\beta = \mathbf{b} \odot [\mathbf{p} + (1 - \mathbf{p}) \odot \tilde{\mathbf{w}}] + (1 - \mathbf{b}) \odot \mathbf{p}, \quad (6)$$

where $\odot$ denotes the Hadamard product, and $\mathbf{p} \in \mathbb{R}^{B \times |\mathcal{V}|}$ represents the predicted probability of each token from patch-level visual representations $\mathbf{X}$. The predicted probability $\mathbf{p}$ provides a measure of prediction

confidence, while $\tilde{\mathbf{w}}$ captures clinical importance. Their integration in β adaptively modulates the contribution of each token to the optimization objective. Positive and negative tokens are subsequently reweighted according to different criteria. For a positive token ($\mathbf{b}_{i,j} = 1$), its weighting factor is expressed as:

$$\beta_{i,j} = \mathbf{p}_{i,j} + (1 - \mathbf{p}_{i,j}) \odot \tilde{\mathbf{w}}_{i,j}. \quad (7)$$

Thus, clinically relevant tokens with low predicted probabilities ($\mathbf{p}_{i,j} \rightarrow 0$) are treated as hard positives and receive larger optimization weights, thereby inducing stronger gradients. Conversely, as a token becomes increasingly well captured by the visual representations ($\mathbf{p}_{i,j} \rightarrow 1$), its weighting factor approaches unity, thereby avoiding excessive optimization of tokens that are already well represented. For a negative token ($\mathbf{b}_{i,j} = 0$), its weighting factor becomes:

$$\beta_{i,j} = \mathbf{p}_{i,j}. \quad (8)$$

This formulation assigns greater weights to hard negatives with high predicted probabilities while down-weighting easy negatives with low predicted probabilities. Such adaptive reweighting enables CheXLoc to focus optimization on clinically informative and visually challenging tokens.

To further enhance optimization stability, we incorporate focal modulation [53]:

$$\mathcal{L}_{i,j} = \beta_{i,j} \cdot (1 - \tilde{\mathbf{p}}_{i,j})^\gamma \cdot \text{BCE}(\mathbf{p}_{i,j}, \mathbf{b}_{i,j}), \quad (9)$$

where $\tilde{\mathbf{p}}_{i,j} = \mathbf{b}_{i,j} \cdot \mathbf{p}_{i,j} + (1 - \mathbf{b}_{i,j}) \cdot (1 - \mathbf{p}_{i,j})$, $\gamma = 2$ controls the focal modulation strength, and $\text{BCE}(\cdot)$ is the binary cross-entropy loss. To prevent optimization from being dominated by numerous uninformative negative tokens, the loss is normalized by the effective number of informative tokens:

$$N_i = \max \left\{ 1, \sum_j [\mathbf{b}_{i,j} + (1 - \mathbf{b}_{i,j}) \cdot \mathbb{I}(\mathbf{p}_{i,j} > \tau)] \right\}, \quad (10)$$

where $\tau = 0.1$ controls the selection of hard negatives. The final token-level alignment objective is:

$$\mathcal{L}_{\text{local}} = \frac{1}{B} \sum_{i=1}^B \frac{1}{N_i} \sum_{j=1}^{|\mathcal{V}|} \mathcal{L}_{i,j}. \quad (11)$$

This objective reduces the influence of abundant easy negatives while maintaining emphasis on clinically relevant tokens and challenging visual-textual associations.

Training objective. The overall pretraining objective jointly optimizes global image-report alignment and entity-guided local patch-token alignment:

$$\mathcal{L}_{\text{pretrain}} = \mathcal{L}_{\text{global}} + \mathcal{L}_{\text{local}}. \quad (12)$$

This objective encourages localized visual representations to emerge from image-report pairs, without requiring spatial annotations.

4.2 Datasets

The full MIMIC-CXR dataset [21] contains 60,440 patients, 246,162 radiographs and 154,482 reports (Fig. 1d). We used its official training split for pretraining, which comprises 59,693 patients, 240,197 radiographs and 150,957 reports (Fig. 3b). To prevent data leakage, no images or reports from the evaluation sets were used during pretraining or downstream adaptation. To assess the generalizability of the learned representations across downstream clinical applications, we evaluated CheXLoc on report generation, disease classification, and lesion segmentation tasks using a diverse collection of public chest X-ray datasets (Fig. 1d). Specifically, report generation was evaluated on the MIMIC-CXR test set and three external datasets—CheXpert Plus [33], ReXGradient [34], and IU X-ray [40]—spanning different institutions, patient populations, and clinical reporting practices. Image classification was evaluated on COVIDx [41], SIIM [42], Shenzhen [43], and TBX11K [44], covering thoracic abnormalities including COVID-19, pneumonia, pneumothorax, and tuberculosis. In addition, we assessed the transferability of the learned visual representations to supervised segmentation tasks on RSNA [45], SIIM [42], and TBX11K [44], which provide lesion-level annotations

for pneumonia, pneumothorax, and tuberculosis, respectively. Detailed dataset statistics and partitioning schemes are provided below.

MIMIC-CXR. MIMIC-CXR [21] is a large-scale publicly available chest radiograph dataset containing radiographic images in JPG format paired with free-text reports. The dataset was collected at Beth Israel Deaconess Medical Center (Boston, MA, USA) between 2011 and 2016. We followed the official data split, comprising 240,197, 2,113, and 3,852 radiographs for training, validation, and testing, respectively. To ensure fair comparison with recent medical multimodal large language models, we additionally evaluated CheXLoc on the ReXRank benchmark [54], which uses a standardized subset of 2,347 test studies drawn from the full MIMIC-CXR test set.

CheXpert Plus. CheXpert Plus [33] is a chest X-ray dataset derived from the Stanford Health Care system and includes radiographs, radiology reports, and de-identified clinical metadata. Following the ReXRank evaluation protocol [54], we excluded studies with empty *Findings* sections and adopted the official partition. The resulting dataset comprises 27,299 patients (83,253 images) for training and 62 patients (102 images) for testing.

ReXGradient. ReXGradient [34] is a multi-institutional chest radiograph dataset comprising paired images and reports collected from three U.S. healthcare systems and 79 institutions. We used the official split, with 140,000 studies (238,968 images) for training, 10,000 studies (17,007 images) for validation, and 10,000 studies (17,029 images) for testing. The training, validation, and test sets span 78, 66, and 67 institutions, respectively.

IU X-ray. IU X-ray [40] is a publicly available chest radiograph dataset released by Indiana University. It contains 2,955 patients with paired radiographic examinations and corresponding radiology reports. Each examination includes frontal and lateral chest radiographs accompanied by free-text clinical descriptions. We evaluated the entire dataset in a zero-shot setting to assess the generalizability of the learned representations to an external clinical domain not observed during pretraining.

COVIDx. COVIDx CXR-3 [41] is an international chest radiograph dataset for COVID-19 diagnosis. It contains 30,386 radiographs from 17,026 patients collected across more than 50 countries, including 16,194 COVID-19-positive and 14,192 negative cases. To ensure a consistent evaluation protocol, we used the same test set of 400 cases for zero-shot disease classification across all methods.

Shenzhen. Shenzhen [43] is a tuberculosis chest radiograph dataset jointly collected by the U.S. National Library of Medicine, Shenzhen Third People’s Hospital, and Guangdong Medical University. Following the CheXWorld protocol [32], we partitioned the dataset into 463, 65, and 134 images for training, validation, and testing, respectively.

SIIM. SIIM [42] is a chest radiograph dataset designed for pneumothorax detection and segmentation. As no official data partition is available, we followed the standardized BenchX protocol [55], resulting in 9,303, 1,372, and 1,372 images for training, validation, and testing, respectively.

TBX11K. TBX11K [44] is a tuberculosis chest radiograph dataset containing both image-level labels and lesion-level annotations. This design enables evaluation of both disease classification and lesion segmentation. Following the BenchX protocol [55], we adopted the standardized split comprising 5,879, 1,260, and 1,260 images for training, validation, and testing, respectively.

RSNA. RSNA [45] is a chest radiograph dataset released through the RSNA Pneumonia Detection Challenge. The dataset provides region-level annotations for pneumonia-related abnormalities, facilitating quantitative evaluation of lesion segmentation performance. Following the BenchX protocol [55], we used 18,678, 4,003, and 4,003 images for training, validation, and testing, respectively.

4.3 Implementation details

CheXLoc was pretrained on the MIMIC-CXR training set. Following PriorRG [13] and RADDINO [31], all chest radiographs were resized to 518×518 pixels and processed using the same data augmentation strategy. The model was optimized with AdamW using an initial learning rate of 5×10^{-5} and a ReduceLROnPlateau scheduler with a patience of two epochs. Pretraining was conducted for 10 epochs with a batch size of 48 on a single NVIDIA RTX 5880 Ada GPU, with a total training time of approximately 30 h. The importance weights associated with anatomy, observation, negation, and temporal entities in α were initialized to 1.5, 2.0, 2.0, and 1.5, respectively.

To assess the transferability of the learned visual representations, the pretrained encoders were adapted

to report generation, disease classification, and lesion segmentation using task-specific optimization protocols (Fig. 2a). For report generation, methods without an integrated language decoder employed DistilGPT-2 [56] as the report generator. The visual encoder and language decoder were jointly optimized using the language-modeling objective. For zero-shot report generation on IU X-ray, models fine-tuned on MIMIC-CXR were directly evaluated without further adaptation.

For disease classification, zero-shot prediction for all methods followed the CLIP paradigm [57]. To ensure a consistent evaluation protocol, we used the same ten textual prompts for each disease across all methods to capture linguistic variation. Each prompt was encoded using the corresponding method-specific text encoder, or the CLIP text encoder for methods without an integrated text encoder. The resulting prompt representations were averaged to obtain a disease-level representation. Image-text similarities were then computed between the image representation and each disease-level representation, and the disease with the highest similarity was assigned as the prediction. For supervised classification, a linear classification layer was added to each pretrained visual encoder to map the extracted visual representations to the target label space. All parameters were jointly optimized using cross-entropy loss. For the 1% and 10% training-data settings, subsets were randomly sampled from the full training set and shared across all methods to ensure identical training data. All methods were trained for 50 epochs using AdamW with ReduceLROnPlateau for learning-rate scheduling. The initial learning rate was set to the default value specified in the official implementation of each method.

For lesion segmentation, the final hidden-state representations of each pretrained visual encoder were passed to a lightweight decoder consisting of convolutional and upsampling layers. The encoder and decoder were optimized jointly using a combination of Dice and Focal losses [58]. All methods were trained for 50 epochs using AdamW with ReduceLROnPlateau for learning-rate scheduling. The initial learning rate was set to the default value specified in the official implementation of each method.

For inference-time evaluation, inference time and peak GPU memory usage were measured on the ReXRANK-standardized MIMIC-CXR test subset ($n = 2,347$) using the same GPU device. As the multimodal foundation models are typically constrained to single-sample inference, all models were evaluated with a batch size of 1 to ensure a fair comparison. Each model was evaluated using the precision, decoding strategy, num_beams, and max_new_tokens specified by its official implementation, without further modification. Inference time was measured over the complete test subset and reported as the mean wall-clock time per sample, including input preprocessing, model inference, and report generation. Peak GPU memory usage was recorded under the same inference protocol. Although CheXLoc supports larger batch sizes when sufficient GPU memory is available, all primary comparisons were performed with a batch size of 1; the corresponding trade-off between batch size, inference time, and GPU memory usage is reported in Supplementary Fig. S3.

4.4 Evaluation metrics

Chest X-ray report generation. Report generation performance was evaluated following the ReXRANK protocol. We report lexical and clinically oriented metrics. Specifically, BLEU [23] was used as the lexical metric, whereas SembScore [24], RadGraph [25], RATEScore [26], GREEN [27], and 1/RadCliQ-V1 [22] were considered as clinically oriented metrics. Higher scores indicate better performance. BLEU-n measures the n-gram precision between generated and reference reports. In this study, BLEU refers specifically to BLEU-2, whereas BLEU-4 is reported separately. SembScore [24] evaluates clinical semantic similarity using pathology representations derived from CheXbert [24]. RadGraph [25] measures agreement in clinical entities and relations extracted from generated and reference reports. RATEScore [26] and GREEN [27] provide radiologist-aligned assessments of clinical report quality and clinically significant errors, respectively. We additionally report 1/RadCliQ-V1, the reciprocal of RadCliQ-V1 [22], as used in the ReXRANK [54] evaluation protocol.

Disease classification and lesion segmentation. Disease classification performance was assessed using the area under the receiver operating characteristic curve (AUROC), which quantifies discriminative performance across decision thresholds. Lesion segmentation performance was evaluated using the mean Dice coefficient, which measures the spatial overlap between predicted and reference lesion regions.

Statistical analysis. The results presented in Fig. 2c were obtained directly from the publicly available ReXRANK leaderboard [54] and were therefore not subjected to statistical testing. For experiments not based on the ReXRANK leaderboard, CheXLoc was compared with each competing method separately within each

evaluation setting using 1,000 paired bootstrap resamples of the test set, with the two methods evaluated on the same resampled cases. The 95% confidence intervals (CIs) were calculated using the 2.5th and 97.5th percentiles of the bootstrap distributions, and two-sided p values were calculated from the centered bootstrap distribution of performance differences. Given the multiple metrics, datasets, models, and diseases considered, these analyses were interpreted as exploratory. Nominal significance thresholds of $p < 0.05$, $p < 0.01$, and $p < 0.001$ were used.

4.5 Comparative methods

We benchmarked CheXLoc against a diverse range of state-of-the-art multimodal foundation models and radiology-specific representation learning methods under standardized evaluation protocols. Model size, pre-training data scale, and training supervision for all comparative methods are summarized in Supplementary Table S1.

For chest X-ray report generation, we considered three groups of comparators: general-purpose multimodal foundation models, represented by GPT-4V [35]; medical-domain multimodal large language models, including MedGemma-4B [36], MedVersa-7B [37], and RadFM-14B [30]; and radiology-specific report generation models, including Libra-3B [28], CheXagent-3B [38], RadPhi3.5Vision-3.8B [39], and MAIRA-2-7B [29]. Performance results for these methods were obtained from the ReXRank benchmark [54], which provides a standardized evaluation framework for chest X-ray report generation.

For chest X-ray report generation, comparators included general-purpose multimodal foundation models (GPT-4V [35]), medical-domain multimodal large language models (MedGemma-4B [36], MedVersa-7B [37], and RadFM-14B [30]), and radiology-specific report generation methods (Libra-3B [28], CheXagent-3B [38], RadPhi3.5Vision-3.8B [39], and MAIRA-2-7B [29]). Performance metrics for these methods were obtained from the ReXRank benchmark [54], which provides a unified and rigorously curated evaluation framework for chest X-ray report generation.

For downstream disease classification and lesion segmentation, we compared CheXLoc with representative radiology representation learning methods, including AFLoc [5], CheXWorld [32], RADDINO [31], EGMA [16], and BiomedCLIP [12]. These methods span vision-language pretraining and radiology-specific representation learning. For consistency and reproducibility, all methods were evaluated using the same datasets, preprocessing procedures, and evaluation metrics.

4.6 Localization-aware evaluation for report generation.

Existing explanation metrics, including sufficiency and comprehensiveness from ERASER [59], insertion and deletion from RISE [60], and smallest sufficient regions [61], were developed primarily for classification models and post hoc explanations. Consequently, these metrics do not directly assess the sensitivity of generated clinical findings to disease-specific image regions. To address this limitation, we introduce Localization-Aware Generation Evaluation (LAGE), a perturbation-based framework that quantifies the dependence of generated findings on lesion-specific image evidence.

For a target lesion c , let x denote the original chest radiograph and M_c the corresponding lesion mask. We construct two complementary perturbations. A lesion-only image retains only the target lesion region,

$$x_c^{\text{only}} = x \odot M_c. \quad (13)$$

whereas a lesion-removed image removes the target lesion while preserving all remaining image content,

$$x_c^{\text{rm}} = x \odot (1 - M_c). \quad (14)$$

Following the CheXbert evaluation protocol [24], each generated report is converted into a binary vector of clinical observations:

$$\hat{\mathbf{y}}_i = (\hat{y}_{i,1}, \dots, \hat{y}_{i,K}) \in \{0, 1\}^K, \quad (15)$$

where K denotes the number of clinical observations, and $\mathbf{y}_i = (y_{i,1}, \dots, y_{i,K}) \in \{0, 1\}^K$ indicates the corresponding reference labels derived from the ground-truth report.

(1) **Sufficiency.** Sufficiency measures whether the isolated lesion alone provides adequate evidence for generating the target finding while suppressing unrelated findings. For a target lesion c , the sufficiency score for the i^{th} sample is defined as:

$$S_i(c) = \mathbb{I}(\hat{y}_{i,c}^{\text{only}} = y_{i,c}) \left(1 - \frac{1}{K-1} \sum_{k \neq c} \hat{y}_{i,k}^{\text{only}} \right), \quad (16)$$

where $\hat{y}_{i,c}^{\text{only}}$ denotes the predicted label of lesion c for the i^{th} sample computed from the lesion-only image. The metric is intentionally designed to reward both recovery of the target finding and suppression of unrelated findings under lesion-only input. The overall sufficiency for lesion c is computed as:

$$\text{Sufficiency}(c) = \frac{1}{N_c} \sum_{i=1}^{N_c} S_i(c), \quad (17)$$

where N_c is the number of samples containing lesion c . A high sufficiency score indicates that the isolated lesion alone supports accurate generation of the target clinical finding while minimizing spurious predictions for unrelated findings.

(2) **Necessity.** Necessity evaluates whether removing a lesion suppresses the target finding while leaving other findings unchanged. Following lesion removal, the label corresponding to the target finding is set to zero, while all other labels are retained. This counterfactual labeling assumes that removing the annotated target lesion eliminates its corresponding clinical finding while leaving other findings unchanged, solely for the purpose of evaluation and without implying clinical causality. Based on this assumption, the counterfactual reference labels are defined as:

$$\tilde{\mathbf{y}}_i = (y_{i,1}, \dots, 0, \dots, y_{i,K}). \quad (18)$$

Necessity is computed as the macro-F1 score between predictions $\hat{\mathbf{y}}_i^{\text{rm}}$ generated from the lesion-removed image and the modified reference labels, denoted as:

$$\text{Necessity}(c) = \frac{1}{N_c} \sum_{i=1}^{N_c} \text{Macro-F1}(\hat{\mathbf{y}}_i^{\text{rm}}, \tilde{\mathbf{y}}_i), \quad (19)$$

(3) **Localization-Aware Generation Evaluation (LAGE).** To jointly evaluate these complementary properties, we define Localization-Aware Generation Evaluation (LAGE) as the arithmetic mean of sufficiency and necessity:

$$\text{LAGE}(c) = \frac{1}{2} (\text{Sufficiency}(c) + \text{Necessity}(c)). \quad (20)$$

LAGE quantifies sensitivity of generated findings to controlled lesion-specific perturbations. It therefore provides evidence of localization awareness rather than establishing causal image-text grounding.

5 Data availability

All datasets used in this study are publicly accessible, subject to the access conditions of their respective repositories. **MIMIC-CXR** is available through PhysioNet under a credentialed-access data use agreement. **CheXpert Plus**, **ReXGradient**, **IU X-ray**, **COVIDx**, **Shenzhen**, **SIIM**, **TBX11K**, and **RSNA** are available from their respective public repositories.

6 Code availability

The code and benchmark will be released at <https://github.com/mk-runner/CheXLoc>. The code used for statistical analysis is available at <https://gist.github.com/rajpurkar/2cc9f61c8f4b0b56a52d5959a47ea7f8>.

References

- [1] Ma, D., Pang, J., Gotway, M. B. & Liang, J. A fully open ai foundation model applied to chest radiography. *Nature* 1–11 (2025).
- [2] Yao, J. et al. Eva-x: A foundation model for general chest x-ray analysis with self-supervised learning. *npj Digital Medicine* **8**, 678 (2025).
- [3] Dong, K., Cheng, Y., He, K. & Suo, J. A generative model uses healthy and diseased image pairs for pixel-level chest x-ray pathology localization. *Nature Biomedical Engineering* 1–13 (2025).
- [4] Lai, H. et al. Carzero: Cross-attention alignment for radiology zero-shot classification. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 11137–11146 (2024).
- [5] Yang, H. et al. A multimodal vision–language model for generalizable annotation-free pathology localization. *Nature Biomedical Engineering* 1–15 (2026).
- [6] Park, J., Yoon, B., Kim, S. & Choi, K. Radzero: Similarity-based cross-attention for explainable vision–language alignment in chest x-ray with zero-shot multi-task capability. In Belgrave, D. et al. (eds.) *Advances in Neural Information Processing Systems*, vol. 38, Main Conference, 56008–56034 (Curran Associates, Inc., 2025). URL https://proceedings.neurips.cc/paper_files/paper/2025/file/510e5c2fd5ad326f75594563f4ad5e0d-Paper-Conference.pdf.
- [7] Liu, K. et al. Seeing like radiologists: Context- and gaze-guided vision-language pretraining for chest x-rays (2026). URL <https://arxiv.org/abs/2603.26049>.
- [8] Zambrano Chaves, J. M. et al. A clinically accessible small multimodal radiology model and evaluation metric for chest x-ray findings. *Nature Communications* **16**, 3108 (2025).
- [9] Tanno, R. et al. Collaboration between clinicians and vision–language models in radiology report generation. *Nature Medicine* **31**, 599–608 (2025).
- [10] Wang, Z., Wu, Z., Agarwal, D. & Sun, J. Medclip: Contrastive learning from unpaired medical images and text. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 3876–3887 (2022).
- [11] Zhang, X., Wu, C., Zhang, Y., Xie, W. & Wang, Y. Knowledge-enhanced visual-language pre-training on chest radiology images. *Nature Communications* **14**, 4542 (2023).
- [12] Zhang, S. et al. A multimodal biomedical foundation model trained from fifteen million image–text pairs. *NEJM AI* **2**, AIoa2400640 (2025). URL <https://ai.nejm.org/doi/full/10.1056/AIoa2400640>.
- [13] Liu, K. et al. Priorrg: Prior-guided contrastive pre-training and coarse-to-fine decoding for chest x-ray report generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, 7206–7214 (2026).
- [14] Liu, K. et al. Enhanced contrastive learning with multi-view longitudinal data for chest x-ray report generation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 10348–10359 (2025).
- [15] Wang, F. & Yu, L. Scaling chest x-ray foundation models from mixed supervisions for dense prediction. *IEEE Transactions on Medical Imaging* 1–1 (2025).
- [16] Ma, C. et al. Eye-gaze guided multi-modal alignment for medical representation learning. In *Advances in Neural Information Processing Systems*, vol. 37, 6126–6153 (2024).
- [17] Wang, F., Zhou, Y., Wang, S., Vardhanabhuti, V. & Yu, L. Multi-granularity cross-modal alignment for generalized medical visual representation learning. In *Advances in Neural Information Processing Systems*, vol. 35, 33536–33549 (2022).

- [18] Zhao, W., Huang, Z., Feng, J. & Wang, X. Superclip: Clip with simple classification supervision. In Belgrave, D. et al. (eds.) *Advances in Neural Information Processing Systems*, vol. 38, Main Conference, 117828–117850 (Curran Associates, Inc., 2025). URL https://proceedings.neurips.cc/paper_files/paper/2025/file/aadaedf7fc275bf68ad217bf1b38626e-Paper-Conference.pdf.
- [19] Dietterich, T. G., Lathrop, R. H. & Lozano-Pérez, T. Solving the multiple instance problem with axis-parallel rectangles. *Artificial Intelligence* **89**, 31–71 (1997).
- [20] Ilse, M., Tomczak, J. & Welling, M. Attention-based deep multiple instance learning. In *Proceedings of the International Conference on Machine Learning*, 2127–2136 (PMLR, 2018).
- [21] Johnson, A. et al. MIMIC-CXR-JPG - chest radiographs with structured labels. *PhysioNet* (2024). URL <https://doi.org/10.13026/jsn5-t979>. Version 2.1.0.
- [22] Yu, F. et al. Evaluating progress in automatic chest x-ray radiology report generation. *Patterns* **4**, 100802 (2023).
- [23] Papineni, K., Roukos, S., Ward, T. & Zhu, W.-J. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, 311–318 (2002).
- [24] Smit, A. et al. Combining automatic labelers and expert annotations for accurate radiology report labeling using BERT. In Webber, B., Cohn, T., He, Y. & Liu, Y. (eds.) *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 1500–1519 (Association for Computational Linguistics, Online, 2020). URL <https://aclanthology.org/2020.emnlp-main.117/>.
- [25] Jain, S. et al. Radgraph: Extracting clinical entities and relations from radiology reports. In Vanschoren, J. & Yeung, S. (eds.) *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, vol. 1 (2021). URL https://datasets-benchmarks-proceedings.neurips.cc/paper_files/paper/2021/file/c8ffe9a587b126f152ed3d89a146b445-Paper-round1.pdf.
- [26] Zhao, W. et al. RaTEScore: A metric for radiology report generation. In Al-Onaizan, Y., Bansal, M. & Chen, Y.-N. (eds.) *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 15004–15019 (Association for Computational Linguistics, Miami, Florida, USA, 2024). URL <https://aclanthology.org/2024.emnlp-main.836/>.
- [27] Ostmeier, S. et al. GREEN: Generative radiology report evaluation and error notation. In Al-Onaizan, Y., Bansal, M. & Chen, Y.-N. (eds.) *Findings of the Association for Computational Linguistics: EMNLP 2024*, 374–390 (Association for Computational Linguistics, Miami, Florida, USA, 2024). URL <https://aclanthology.org/2024.findings-emnlp.21/>.
- [28] Zhang, X., Meng, Z., Lever, J. & Ho, E. S. L. Libra: Leveraging temporal images for biomedical radiology analysis. In *Findings of the Association for Computational Linguistics: ACL 2025*, 17275–17303 (Association for Computational Linguistics, Vienna, Austria, 2025).
- [29] Bannur, S. et al. Maira-2: Grounded radiology report generation (2024). [2406.04449](https://arxiv.org/abs/2406.04449).
- [30] Wu, C. et al. Towards generalist foundation model for radiology by leveraging web-scale 2d&3d medical data. *Nature Communications* **16**, 7866 (2025).
- [31] Perez-Garcia, F. et al. Exploring scalable medical image encoders beyond text supervision. *Nature Machine Intelligence* **7**, 119–130 (2025).

- [32] Yue, Y. et al. Chexworld: Exploring image world modeling for radiograph representation learning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 20778–20788 (2025).
- [33] Chambon, P. et al. Chexpert plus: Augmenting a large chest x-ray dataset with text radiology reports, patient demographics and additional image formats (2024). [2405.19538](#).
- [34] Zhang, X., Acosta, J. N., Miller, J., Huang, O. & Rajpurkar, P. ReXGradient-160k: A large-scale publicly available dataset of chest radiographs with free-text reports. In *Machine Learning for Health 2025* (2025). URL <https://openreview.net/forum?id=gujOcWR0vN>.
- [35] OpenAI. Gpt-4v(ision) system card. Technical Report, OpenAI (2023).
- [36] Sellergren, A. et al. Medgemma technical report (2026). URL <https://arxiv.org/abs/2507.05201>.
- [37] Zhou, H.-Y. et al. Medversa: a generalist foundation model for diverse medical imaging tasks. *NEJM AI* **3**, A10a2500595 (2026).
- [38] Chen, Z. et al. Chexagent: Towards a foundation model for chest x-ray interpretation. In *AAAI 2024 Spring Symposium on Clinical Foundation Models* (2024). URL <https://openreview.net/forum?id=P3LOmrZWGR>.
- [39] Ranjit, M., Srivastav, S. & Ganu, T. Radphi-3: Small language models for radiology (2024). URL <https://arxiv.org/abs/2411.13604>. [2411.13604](#).
- [40] Demner-Fushman, D. et al. Preparing a collection of radiology examinations for distribution and retrieval. *Journal of the American Medical Informatics Association* **23**, 304–310 (2016).
- [41] Pavlova, M. et al. Covidx cxr-3: A large-scale, open-source benchmark dataset of chest x-ray images for computer-aided covid-19 diagnostics (2022). URL <https://arxiv.org/abs/2206.03671>. [2206.03671](#).
- [42] Zawacki, A. et al. Siim-acr pneumothorax segmentation. <https://kaggle.com/competitions/siim-acr-pneumothorax-segmentation> (2019). Kaggle.
- [43] Jaeger, S. et al. Automatic tuberculosis screening using chest radiographs. *IEEE Transactions on Medical Imaging* **33**, 233–245 (2013).
- [44] Liu, Y., Wu, Y.-H., Ban, Y., Wang, H. & Cheng, M.-M. Rethinking computer-aided tuberculosis diagnosis. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2643–2652 (2020).
- [45] Shih, G. et al. Augmenting the national institutes of health chest radiograph dataset with expert annotations of possible pneumonia. *Radiology: Artificial Intelligence* **1**, e180041 (2019).
- [46] Boecking, B. et al. MS-CXR: Making the Most of Text Semantics to Improve Biomedical Vision-Language Processing. *PhysioNet* (2024). URL <https://doi.org/10.13026/9g2z-jg61>. Version 1.1.0.
- [47] Zhou, B., Khosla, A., Lapedriza, A., Oliva, A. & Torralba, A. Learning deep features for discriminative localization. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2921–2929 (2016).
- [48] Zhang, D., Han, J., Cheng, G. & Yang, M.-H. Weakly supervised object localization and detection: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **44**, 5866–5885 (2022).
- [49] Boecking, B. et al. Making the most of text semantics to improve biomedical vision-language processing. In *Proceedings of the European Conference on Computer Vision*, 1–21 (Springer, 2022).
- [50] Dosovitskiy, A. et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *Proceedings of the International Conference on Learning Representations* (OpenReview.net, 2021). URL <https://openreview.net/forum?id=YicbFdNTTy>.

- [51] Jaegle, A. et al. Perceiver: General perception with iterative attention. In *Proceedings of the International Conference on Machine Learning*, vol. 139, 4651–4664 (PMLR, 2021).
- [52] Tian, Y., Fan, L., Isola, P., Chang, H. & Krishnan, D. Stablerep: Synthetic images from text-to-image models make strong visual representation learners. In *Advances in Neural Information Processing Systems*, vol. 36 (2024).
- [53] Lin, T.-Y., Goyal, P., Girshick, R., He, K. & Dollár, P. Focal loss for dense object detection. In *Proceedings of the International Conference on Computer Vision*, 2980–2988 (2017).
- [54] Zhang, X. et al. Rexrank: A public leaderboard for ai-powered radiology report generation. In Wu, J., Zhu, J., Xu, M. & Jin, Y. (eds.) *Proceedings of The First AAAI Bridge Program on AI for Medicine and Healthcare*, vol. 281 of *Proceedings of Machine Learning Research*, 90–99 (PMLR, 2025). URL <https://proceedings.mlr.press/v281/zhang25b.html>.
- [55] Zhou, Y. et al. Benchx: A unified benchmark framework for medical vision-language pretraining on chest x-rays. In *Advances in Neural Information Processing Systems*, vol. 37, 6625–6647 (2024).
- [56] Sanh, V., Debut, L., Chaumond, J. & Wolf, T. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. In *NeurIPS EMC² Workshop* (2019).
- [57] Radford, A. et al. Learning transferable visual models from natural language supervision. In *Proceedings of the International Conference on Machine Learning*, 8748–8763 (2021).
- [58] Ma, J. et al. Loss odyssey in medical image segmentation. *Medical Image Analysis* **71**, 102035 (2021).
- [59] DeYoung, J. et al. ERASER: A benchmark to evaluate rationalized NLP models. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4443–4458 (Association for Computational Linguistics, Online, 2020). URL <https://aclanthology.org/2020.acl-main.408/>.
- [60] Petsiuk, V., Das, A. & Saenko, K. RISE: randomized input sampling for explanation of black-box models. In *British Machine Vision Conference 2018, BMVC 2018, Newcastle, UK, September 3-6, 2018*, 151 (BMVA Press, 2018). URL <http://bmvc2018.org/contents/papers/1064.pdf>.
- [61] Dabkowski, P. & Gal, Y. Real time image saliency for black box classifiers. In Guyon, I. et al. (eds.) *Advances in Neural Information Processing Systems*, vol. 30 (Curran Associates, Inc., 2017). URL https://proceedings.neurips.cc/paper_files/paper/2017/file/0060ef47b12160b9198302ebdb144dcf-Paper.pdf.

Acknowledgements

This work is supported by the National Natural Science Foundation of China (No. 62272364), the provincial Key Research and Development Program of Shaanxi (No. 2024GH-ZDXM-47), and the Higher Education Science Research Planning Project of China Association of Higher Education (No. 24PG0101).

Author contributions

Kang Liu and Zhuoqi Ma conceptualized the study, developed the methodology, and performed the software implementation, experiments, and formal analysis. Kun Xie, Yunan Li, Ruyi Liu, and Ronghan Li contributed to experimental evaluation and analysis. Guang Jia contributed to study design and methodological development. Kang Liu and Zhuoqi Ma wrote the original manuscript. Qiguang Miao provided supervision, contributed to study design and methodological development, and acquired funding. All authors contributed to interpretation of the results, reviewed and revised the manuscript, and approved the final version.

835 **Competing interests**

836 The authors declare no competing interests.

Table S1: **Comparison of multimodal foundation models and radiology-specific representation learning methods.** #Param denotes the total number of parameters. For AFLoc, RADDINO, CheXWorld, EGMA, BiomedCLIP, and CheXLoc, the reported parameter count includes the additional DistilGPT2 text generator used for report generation.

Method	#Param	Resolution	Pretraining data	Training supervision	Data access	Data sources
AFLoc	231M	299×299	220,000	image-report	public	MIMIC-CXR, Retinal Fundus, Quilt-1M
RADDINO	183M	518×518	838,336	image-only	public, private	MIMIC-CXR, CheXpert, BARX, NIH-CXR, PadChest, private data
CheXWorld	182M	224×224	~500,000	image-only	public	MIMIC-CXR, NIH-CXR, CheXpert
BiomedCLIP	292M	224×224	15,282,336	image-report	public	PMC-15M
EGMA	108M	224×224	3,689	image-report, eye-tracking	public	MIMIC-CXR, MIMIC-Eye
Libra-3B	3B	518×518	1,213,097	image-report	public	MIMIC-CXR, MIMIC-Ext-MIMIC-CXR-VQA, Medical-Diff-VQA
CheXagent-3B	3B	512×512	1,052,257	image-report	public	MIMIC-CXR, CheXpert, RSNA, etc. (CheX-Instruct)
MedGemma-4B	4B	448×448	33,139,578	image-report	public	MIMIC-CXR, VQA-RAD, etc.
MedVersa-7B	7B	224×224	~29,000,000	image-report, boxes, masks	public	MIMIC-CXR, CheXpert, etc. (MedInterp)
MAIRA-2-7B	7B	518×518	501,825	image-report, boxes	public, private	MIMIC-CXR, MS-CXR, PadChest, private data
RadFM-14B	14B	512×512	~16,000,000	image-report	public	MedMD
CheXLoc	282M	518×518	240,197	image-report	public	MIMIC-CXR training set

S1 Comparison of representative medical foundation models

Table S1 summarizes representative medical foundation models and radiology-specific representation learning methods considered in this study, including their model size, input resolution, pretraining data scale, training supervision, data accessibility, and major pretraining data sources. For methods originally developed without a report-generation component, a DistilGPT2 text generator was additionally introduced to enable direct comparison in report generation.

S2 Lesion segmentation under limited downstream supervision

To further assess the transferability of the learned visual representations under limited downstream supervision, we evaluated lesion segmentation using 1% and 10% of the training data on the RSNA and TBX11K datasets (Supplementary Table S2). On RSNA, CheXLoc achieved Dice scores of 0.716 and 0.782 with 1% and 10% of the training data, respectively. Although CheXLoc was lower than CheXWorld at 1% supervision (0.775), its performance increased with additional supervision and was comparable to CheXWorld at 10% (0.782 versus 0.776). On TBX11K, CheXLoc achieved Dice scores of 0.910 and 0.939 at 1% and 10% supervision, respectively, yielding the highest numerical performance among the evaluated methods at both supervision levels. These results indicate that the visual representations learned by CheXLoc remain trans-

Table S2: Lesion segmentation under limited downstream supervision. TDF denotes the fraction of the training data used for downstream training. Results are reported as means with 95% confidence intervals estimated from 1,000 bootstrap samples.

Dataset	TDF	AFLoc	RADDINO	CheXWorld	BiomedCLIP	EGMA	CheXLoc
RSNA	1%	0.687 _[0.674, 0.701]	0.518 _[0.504, 0.533]	0.775 _[0.763, 0.787]	0.733 _[0.719, 0.746]	0.475 _[0.462, 0.490]	0.716 _[0.703, 0.729]
RSNA	10%	0.626 _[0.612, 0.640]	0.717 _[0.705, 0.731]	0.776 _[0.764, 0.790]	0.749 _[0.737, 0.762]	0.728 _[0.716, 0.741]	0.782 _[0.771, 0.794]
TBX11K	1%	0.018 _[0.014, 0.022]	0.773 _[0.749, 0.795]	0.905 _[0.888, 0.921]	0.216 _[0.196, 0.237]	0.031 _[0.025, 0.038]	0.910 _[0.896, 0.925]
TBX11K	10%	0.879 _[0.862, 0.895]	0.896 _[0.881, 0.911]	0.905 _[0.889, 0.921]	0.926 _[0.914, 0.939]	0.906 _[0.892, 0.920]	0.939 _[0.928, 0.949]

Table S3: Inference efficiency of CheXLoc across batch sizes for report generation on the ReXRank-standardized MIMIC-CXR test subset ($n = 2,347$).

Batch size	Peak GPU memory (GB)	Average inference time (s/sample)
1	4.789	1.109
2	5.092	0.662
4	5.698	0.416
8	6.912	0.273
16	9.337	0.246
32	14.187	0.263
64	23.163	0.264

ferable to lesion segmentation when downstream annotations are substantially limited, although the relative performance varies across datasets and supervision levels.

S3 Inference efficiency of CheXLoc across batch sizes

Increasing the batch size reduced the average inference time per sample from 1.109 s at batch size 1 to 0.246 s at batch size 16, while peak GPU memory increased from 4.789 to 9.337 GB. Further increases in batch size substantially increased memory usage with little additional reduction in inference time, with batch sizes of 8 and 16 achieving a favourable balance between inference time and GPU memory usage.

S4 Textual prompts for zero-shot disease classification

The textual prompts used for zero-shot disease classification are provided in Table S4. The same prompts were used consistently across all evaluated methods.

Table S4: Textual prompts used for zero-shot disease classification across all evaluated methods.

Disease	Textual Prompt
Pneumonia	1. A chest X-ray showing pneumonia. 2. A chest X-ray showing lung consolidation and infiltrates. 3. A chest radiograph with patchy opacities consistent with pneumonia. 4. Chest X-ray demonstrating alveolar consolidation typical of bacterial pneumonia. 5. A chest X-ray with signs of infection in both lungs. 6. A chest X-ray showing inflammation and airspace disease. 7. Radiograph showing bilateral consolidation due to pneumonia. 8. A chest X-ray indicating acute lower respiratory infection. 9. An X-ray image showing pneumonia with dense opacities. 10. Chest X-ray with pulmonary infiltrates characteristic of pneumonia.
Pneumothorax	1. A chest X-ray showing pneumothorax. 2. Chest radiograph demonstrating collapsed right lung. 3. A chest X-ray with visible pleural line and absent lung markings. 4. A chest X-ray showing air in the pleural space. 5. Radiograph showing left-sided pneumothorax. 6. A chest X-ray showing partial lung collapse. 7. Chest X-ray showing hyperlucent hemithorax due to pneumothorax. 8. A chest X-ray demonstrating spontaneous pneumothorax. 9. Chest radiograph showing absence of vascular markings in one lung field. 10. X-ray image with signs of pneumothorax and pleural separation.
Tuberculosis	1. A chest X-ray showing tuberculosis. 2. A chest radiograph with cavitary lesions in the upper lobes. 3. Chest X-ray showing fibrotic changes and nodular opacities consistent with tuberculosis. 4. A chest X-ray showing patchy infiltrates in both lungs due to tuberculosis. 5. Radiograph showing consolidation and cavity formation typical of pulmonary tuberculosis. 6. Chest X-ray demonstrating apical opacities caused by tuberculosis infection. 7. A chest X-ray showing scarring and nodules consistent with old tuberculosis. 8. Chest radiograph showing upper lobe lesions with cavitation. 9. A chest X-ray indicating pulmonary tuberculosis. 10. X-ray showing lung cavitation and consolidation consistent with tuberculosis.
Normal	1. A normal chest X-ray. 2. A chest X-ray showing clear lungs and no abnormal findings. 3. Chest radiograph without any signs of disease. 4. A chest X-ray with normal appearance and no consolidation. 5. Radiograph demonstrating no acute cardiopulmonary abnormality. 6. Chest X-ray showing no evidence of infection or collapse. 7. A chest X-ray showing healthy lungs. 8. X-ray image without opacity or lesion. 9. A normal chest radiograph with clear lung fields. 10. A chest X-ray showing no pneumothorax, no pneumonia, no tuberculosis.